

A novel method of creating machine learning-based time series meta-models for building energy analysis

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ARTICLE INFO

Article history:

Received 19 October 2022

Revised 8 December 2022

Accepted 27 December 2022

Available online 29 December 2022

Keywords:

Building energy

Meta-model

Distributed lag model

Machine learning

Time series model

ABSTRACT

The *meta-models* have been widely used to replace computationally expensive engineering-based building energy models for model calibration, sensitivity analysis, and performance optimization of buildings. However, most *meta-models* are used for monthly or annual building energy analysis applying the same procedure as building energy or load prediction. A few studies thoroughly explore the characteristics of *meta-models* when creating hourly machine learning-based *meta-models* for building energy assessment, which is very different from conventional building load or energy prediction. Therefore, this paper proposes a new method of creating time series models for building energy analysis based on machine learning techniques. The main feature of this new method is to include two steps (model preselection and data folding) by taking advantage of the sequential nature during the process of constructing *meta-models*. A case study of office buildings is used to demonstrate the application of this new approach using the EnergyPlus program and R computational environment. The results indicate that the time-series *meta-models* constructed using this new method have very high accuracy with a moderate computational cost. For the best models, the R^2 values are above 0.99 and the CV(RMSE) (coefficient variation of root mean square error) values are below 0.06 for both heating and cooling energy prediction. The ensemble machine learning models, including Cubist, gradient boosting machine, and stacking, are recommended to create the *meta-models* for hourly building energy analysis. The distributed lag models combined with these ensemble models can be used to utilize the time-series features of hourly building energy assessment. The method proposed here can be extended to the time-series energy analysis in different time scales (sub-hourly, hourly, or daily) for other energy systems (such as PV, solar thermal, and wind).

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Abbreviations: ABC, approximate Bayesian computation; ADL, autoregressive distributed lag; ANN, artificial neural network; ASHRAE, American Society of Heating, Refrigerating, and Air Conditioning Engineers; CB, cubist; CSP, cooling set-point; CSWD, China Standard Weather Data; CV(RMSE), root mean square error coefficient of variation; DFS, diffuse solar radiation; DL, distributed lag; DNN, deep neural network; DRS, direct solar radiation; DT, day type; EPD, equipment power density; EUR, equipment utilization rate; EWU, exterior wall U-value; FU, floor U-value; GBM, gradient boosting machine; HSP, heating set-point; HVAC, heating, ventilation, and air conditioning; INF, infiltration rate; LHS, Latin hypercube sampling; LPD, lighting power density; LST, light switch time; LSTM, long short-term memory; MAE, mean absolute error; MARS, multivariate adaptive regression splines; MLR, multiple linear regression; PO, people occupancy; R^2 , coefficient of determination; RF, random forest; RH, relative humidity; RMSE, root mean square error; RU, roof U-value; SHGC, solar heat gain coefficient; SVM, support vector machine; Ta, outdoor dry bulb temperature; WS, wind speed; WU, window U-value.

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1. Introduction

The building sector accounts for one-third of total global final energy and approximately 15 % of direct carbon emissions [1]. Hence, it is necessary to reduce energy and associated emissions from buildings. The construction of building energy models can provide insight into the building energy's characteristics and assess retrofit energy savings or carbon emission reduction in buildings. The building energy models can be divided into the forward and data-driven approaches [2]. The forward approach is to estimate energy performance based on the known parameters and inputs by using engineering-based models [3,4], such as EnergyPlus, DeST, and DOE-2. In contrast, the data-driven approach can create a mathematical model based on the known inputs and known outputs by using a number of regression methods, such as multiple

linear regression, neural network, and auto-regressive moving average model [5,6].

The *meta*-models (also named surrogate models, or emulators) can combine both forward and data-driven methods to create fast-run mathematical models by replacing the engineering-based models [7]. The *meta*-models are of particular interest where expensive physics-based simulations for building energy analysis are required for a large number of simulation runs or fast computation. The *meta*-models can approximate complex relationships among inputs and outputs in dynamic physics-based energy models of buildings. A good balance should be obtained between computational cost and model accuracy for *meta*-models. In the field of building energy analysis, the *meta*-models can be used for uncertainty analysis [8], model calibration [9], sensitivity analysis [10], performance optimization [11], and model predictive control [12]. For two-dimensional Monte Carlo uncertainty analysis, the *meta*-models should be used to replace computational expensive energy models since a large number of simulation runs are involved (over millions of simulations in [8]). The *meta*-model is a common method to reduce computational cost in model calibration since there are over 100,000 simulation runs required to reach the convergence of MCMC results [13]. For the variance-based global sensitivity analysis, hundreds of thousands of simulation runs are required to reach the convergence of sensitivity indicators, including total and main effects in building energy assessment [14]. Hence, the *meta*-models can be used to expedite sensitivity computation by replacing the models created from the EnergyPlus, DOE-2 or other building energy simulation programs. Different types of surrogate models (named as black-box or data-driven models) instead of physics-based models can be used to simulate the dynamic characteristic of building performance when applying model predictive control techniques in buildings [12,15]. A good overview of application of *meta*-model in building energy assessment is available in [7].

Machine learning techniques have been widely used to create *meta*-models for building energy analysis [16–19]. Liu et al. [20] create the MARS models based on the EnergyPlus simulation results for campus buildings located in Tianjin, China. Then the variance-based and predictive variable importance approaches are implemented to determine the important factors affecting annual carbon emissions in these campus buildings. Zhang et al. [21] construct the hourly surrogate models for building stock using the EnergyPlus simulation engine. The predictive accuracy of surrogate models obtained is within the ASHRAE guideline for model performance [22]. Chegari et al. [23] develop surrogate models for annual thermal energy demand and discomfort degree-hours for the optimization of a nearly zero energy building. The *meta*-models used are multilayer feed-forward networks in which a sigmoid function for the hidden layer and a linear function for the output layer were used as transfer functions. Li et al. [24] explore the performance of multi-output learning models at daily, monthly, and annual time scales using the Bayesian adaptive spline surface and deep neural network algorithms for an office building located in Tianjin, China. The results indicate that the multi-output models coupled with principal component analysis can estimate accurate energy use at various time scales by maintaining output correlation among outputs. Westermann et al. [25] develop the deep temporal convolutional neural network surrogate models for a small H-shaped office building in the multiple climate zones of Canada. The output variables include annual heating, annual cooling, and hourly heating demand. Zhu et al. [26] develop the Cubist, MARS, and SVM machine learning models for annual and monthly electricity and gas use in an office building. These machine learning models are used for sensitivity analysis and approximate Bayesian computation in the uncertainty calibration

of building energy models. For more research on machine learning-based building energy analysis, please refer to [27–29].

Most of the previous studies are focused on the construction of conventional time series building energy models based on the measurement data [30,31]. For *meta*-models, previous studies are concentrated on monthly or annual building energy use [7] with little discussion of the characteristics of *meta*-models. There are only a few studies on the detailed discussion of creating the hourly *meta*-model analysis of building energy by making full use of the features of the *meta*-model procedure.

Therefore, the aim of this paper would propose a novel method to create the time-series *meta*-models of a building based on machine learning. An office building is used as a case study to demonstrate the characteristics of this new method. The output variables are hourly heating and cooling energy obtained from the engineering-based dynamic building energy models. The method proposed would be easily extended to different time scales, including sub-hourly and daily energy analyses of buildings. The method would provide an alternative method to create time-series *meta*-models for other renewable energy systems, including photovoltaics, solar thermal system, and wind electricity.

The originality of this research lies in proposing a systemically novel two-step method of creating machine learning-based time series *meta*-models for building energy analysis. This method can preselect the good performance models from many machine learning models by first using a small number of data sets. The features of the time series analysis of building energy can be considered by taking into account the lag effect and the autoregressive method combined with the machine learning approaches. Then the predictive performance of selected models can be properly assessed by using the data folding technique. This method can make full use of the characteristics of *meta*-models to sequentially create more training data, which is significantly different from the conventional building energy or load predictions.

2. Method

The procedures of a novel method of creating the time-series *meta*-models of building energy can be divided into three steps as illustrated in Fig. 1. The first step is the data preparation to collocate the data from the engineering-based building energy models. This step aims to construct the data more suitable for time series analysis of building energy. The second step is the model preselection to create a number of machine learning models by considering the characteristics of the time series of building energy and using a small number of data sets. Three types of models are lag-free, distributed lag, and autoregressive distributed lag models. This step aims to select several machine learning-based time series models with good predictive performance for a given data set. The third step is to apply the data folding technique to assess the predictive performance of the models obtained from the previous step using more data created from the engineering-based building energy models. More importantly, the stability of model performance can be evaluated based on this data folding technique. Detailed descriptions of this method are presented in the following sections.

2.1. Data preparation

Fig. 2 shows a four-storey rectangular office building with a total floor area of 3200 m² and a window-wall ratio of 0.4. The thermal properties of the building envelope are compatible with the latest standards of non-domestic buildings in China [32,33]. The internal heat gains and their hourly schedules are also obtained from these two standards [32,33]. A fan coil system is

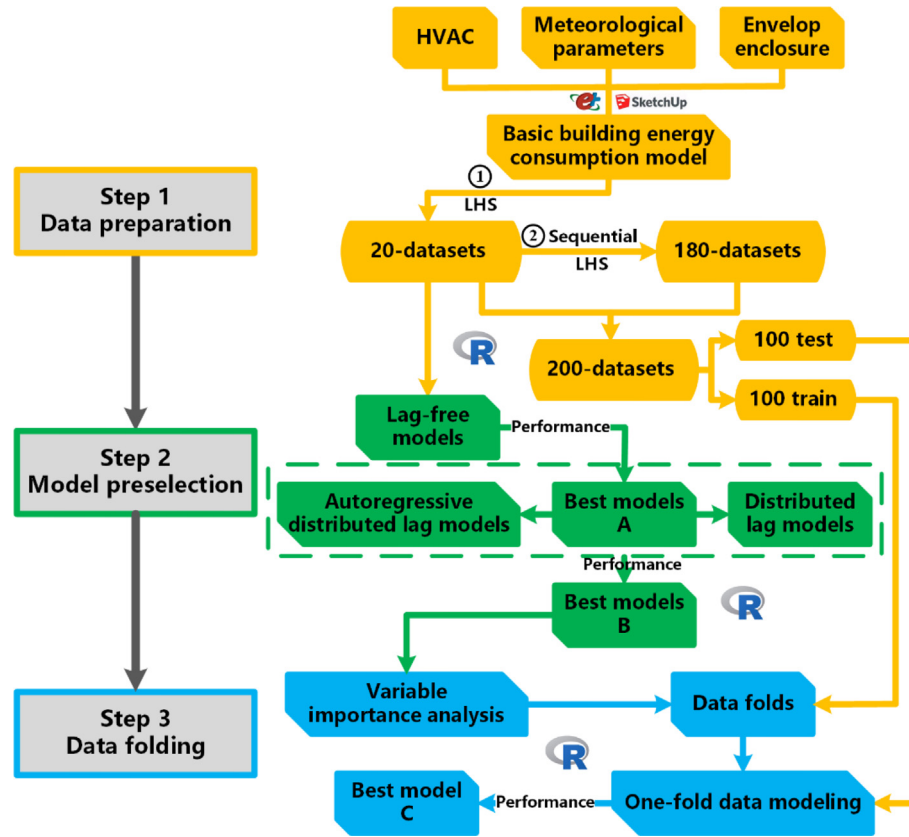


Fig. 1. The flow chart of creating machine learning-based time series *meta*-models for building energy analysis.

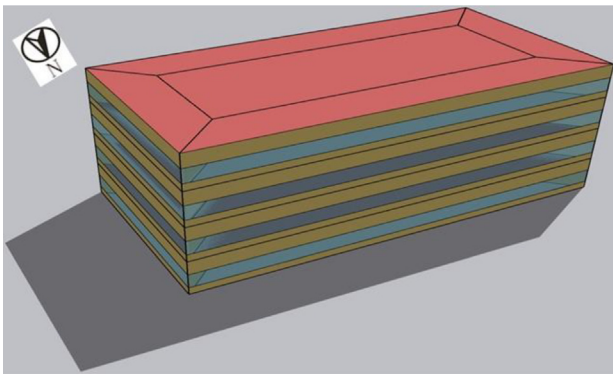


Fig. 2. A four-story office building in Tianjin (China) used in this case study.

used to provide heating, cooling, and ventilation in the office building. A gas boiler and a water-cooled chiller are used to provide hot

water and cold water, respectively, in the building. Table 1 and Table 2 list other parameters used in this research. The building is located in Tianjin, China, and the corresponding weather data is the CSWD (China Standard Weather Data) developed by China Meteorological Bureau and Tsinghua University [34]. Tianjin has a temperature monsoon climate with an annual temperature of 12.9 °C. The 99 % heating design temperature is −8.6 °C and the 1 % cooling design temperature is 29.7 °C. The energy simulation is carried out by using the EnergyPlus V22.1 program [35]. The zoning method is based on the four perimeter zones and one core zones on each level of the building. As a result, there are 20 thermal zones in the building energy simulation model. The hourly heating and cooling energy data are used for two output variables in this research.

The input variables can be divided into four types: statics, dynamic, lagged, and autoregressive variables. The statics variables as listed in Table 1 are the fixed values in annual building energy analysis, which are not changed with monthly, daily, or hourly

Table 1
Building static parameters for time series analysis.

No.	Parameters	Short names	Range	Unit
1	Exterior wall U-value	EWU	0.15–0.3	W/(m ² K)
2	Roof U-value	RU	0.1–0.3	W/(m ² K)
3	Window U-value	WU	0.5–1.5	W/(m ² K)
4	Floor U-value	FU	0.25–0.4	W/(m ² K)
5	Solar heat gain coefficient	SHGC	0.2–0.7	–
6	Equipment power density	EPD	10–15	W/m ²
7	Lighting power density	LPD	7–15	W/m ²
8	Infiltration rate	INF	0.5–0.8	ACH
9	Cooling set-point	CSP	23–26	°C
10	Heating set-point	HSP	20–23	°C

Table 2
Building dynamic parameters for time series analysis.

No.	Parameters	Short names	Time scale	Unit
1	Outdoor dry bulb temperature	Ta	Hour	°C
2	Relative Humidity	RH	Hour	%
3	Wind speed	WS	Hour	m/s
4	Diffuse solar radiation	DFS	Hour	W/m ²
5	Direct Solar Radiation	DRS	Hour	W/m ²
6	People occupancy	PO	Hour	%
7	Equipment utilization rate	EUR	Hour	%
8	Light switch time	LST	Hour	%
9	Hour	Hour	Hour	–
10	Day type	DT	Day	–

time scales. These static variables include the thermal properties of the building envelope, the peak of internal heat gains, and thermal comfort conditions. The ranges of these values are derived from previous studies [20,26] and standards [32,33]. The dynamic variables are the variables that would change with time in building energy analysis as summarized in Table 2. The first five variables are weather conditions, which have significant influences on building energy. The next three variables are the percentage of change in internal heat gains, including occupancy, equipment utilization, and lighting. The last two variables are the time features extracted from the time stamp of energy data, including the hour and day type. The day type denotes the day of the week. This value would be specified as the 1 for Sunday and the 2 for Monday. The subsequent days would follow the same rules till Saturday is taken as 7. The third type of variables are the lagged predictors derived from the dynamic variables listed in Table 2. For instance, the Ta_L0 denotes the outdoor dry bulb temperature in the current hour and the Ta_L1 denotes the outdoor dry bulb temperature in the previous hour. The maximum time lag of 4 is chosen based on the preliminary analysis and other studies [21]. Note that the lags of day type are not considered since the day-lag effect is not expected in the hour-scale of building energy analysis. The fourth type of variable is energy use in the autoregressive model. The maximum lag of building energy is 4-hour based on the preliminary analysis. More detailed descriptions of the lagged variables and associated time series models would be described in sections 2.2.2 and 2.2.3.

The Latin hypercube sampling (LHS) method as an efficient stratified sampling method [36] is used to obtain the input matrix. The first step uses LHS to obtain 20 sets of data for model preselection. The second step is to perform sequential LHS based on the previous samplings to obtain an additional 180 sets of data, in which 100 sets are used for data folding and 100 sets are used for model testing, as will be described in sections 2.2 and 2.3. Then, these static inputs as listed in Table 1 would be imported into the EnergyPlus models to run models to obtain hourly heating and cooling energy of buildings. Every typical year has 8760 h and the corresponding dynamic parameters as dynamic inputs can be also obtained as listed in Table 2. There are 20 inputs (10 static and 10 dynamic variables) and 2 outputs (heating/cooling). The HVAC system only works between 8:00–18:00 on weekdays (11 h per day). The heating system would be turned on for 103 working days per year and the cooling system would be turned on for 86 working days per year. As a result, a $1,133 \times 22$ matrix for heating energy and a 943×22 matrix for cooling energy, respectively, can be obtained per year in this case study. These data sets can be used for creating the conventional machine learning models, i.e. the lag free models. For the distributed lag and autoregressive distributed lag models, the inputs would be increased accordingly, which will be described in sections 2.2.2 and 2.2.3.

2.2. Model preselection

This step is to select the *meta*-models from a number of learning models using a small amount of data. The specific data size used in this step depends on several factors, including accuracy requirement, time limit, and project cost. The 20 annual data sets are used in this case study based on a preliminary study. Three types of time series models are considered, including lag-free, distributed-lag, and autoregressive distributed lag models. These three types of models would be described in detail in the following three subsections.

2.2.1. Lag-free models

The lag-free models refer to the conventional machine learning models without considering the lagged predictors or responses. The predictors in the lag-free models include 10 static variables as listed in Table 1 and 10 dynamic variables as listed in Table 2. The detailed descriptions of these 20 predictors have been described in section 2.1. The lag-free hourly models used in this study can predict 8760 hourly heating and cooling energy use for every combination of 10 static predictors. Hence, the lag-free models are different from the conventional 1-hour or 24-hour ahead models because there is no energy use measured at the initial point for the lag-free models.

The ten machine learning models are used for model selection based on the previous studies [26,37]. These ten models can be divided into three types. The first type of models are three conventional learning models, including MLR (multiple linear regression), MARS (multivariate adaptive regression splines), and SVM (support vector machine). The second type of models are based on neural networks, including ANN (artificial neural network), DNN (deep neural network), and LSTM (long short-term memory). The third type of models are ensemble models, including RF (random forest), GBM, CB (cubist), and stacking models. The DNN model is created using the R H2O environment [38]. The LSTM model is created using the R keras package [39]. The remaining learning models are created using the R caret package [40]. The tuning parameters are determined using the 10-fold cross-validation by considering the bias-variance trade-off [41].

(1) The multiple linear regression model (MLR) is the simplest model using the least square algorithm. The full linear models without the shrinkage methods (such as ridge and step-wise regression) are used in this case study.

(2) The MARS model is an extension of the stepwise linear model by using piecewise linear basis functions. Then these basis functions are used as input predictors instead of original inputs. Two parameters should be tuned: nprune and degree. The nprune refers to the maximum number of terms (including the intercept) as an upper bound of model size. The degree is the maximum degree of interaction. If the degree is 1, the models are simplified

as an additive model.

(3) The SVM model is similar to the robust regression that would be less affected by outliers. Two tuning parameters are cost and sigma. The cost is the main tuning parameter to change the model complexity by penalizing large residuals. A larger cost parameter denotes a more flexible model (likely to over-fit), whereas a smaller cost parameter denotes a more that will more likely to under-fit and less likely to over-fit. The sigma is the tuning parameter in the radial basis function kernel used in this SVM regression.

(4) The ANN model used here is a simple feed-forward neural network with a single hidden layer. There are two parameters to be tuned: size and decay. The size is the number of hidden units in the single hidden layer. The decay is to regularize the model for moderating the over-fitting. As the weight decay increases, the regression model becomes less likely to be overfitting.

(5) The DNN used here is one of the deep neural networks to refer to a multi-layer feedforward artificial neural network with stochastic gradient descent using back propagation. Note that the ANN model used in this paper refers to a classical neural network with three layers (input, hidden, and output), while the DNN model used here belongs to the deep neural networks with a large number of hidden layers. The DNN model is obtained in the H2O environment, an open-source and scalable artificial intelligence platform. There are a large number of parameters that can be tuned in the DNN model, including adaptive rate, activation function, hidden layer & unit, L1 regularization, L2 regularization, and weight decay.

(6) The LSTM model is one of the recurrent neural networks in which the inputs are a sequence, such as time-series data. Hence, the LSTM is suitable for the construction of hourly building energy models. The LSTM models used in this section are regarded as one of lag-free models since the input variables in previous hours are not taken into account. The LSTM models with the previous input variables will be considered in sections 2.2.2 and 2.2.3 as two types of distributed lag models. The LSTM can avoid the vanishing gradient problem from the traditional recurrent neural networks since the LSTM allows gradients to flow unchanged. LSTM includes dense layers and LSTM layer. The dense layer only accepts and processes the simple vector data in the two-dimensional tensor, which is called the fully connected layer. The LSTM layer is a recurrence layer responsible for processing sequence data in a three-dimensional tensor. The hyperparameters include the number of units, activation function, and epochs. A hidden unit is an artificial neuron of a hidden layer and the number of units is the number of these hidden units in each layer of a deep learning model. The activation function in a neural network describes how the weighted sum of the inputs could be converted into outputs in a layer of the neural network. The epoch is the number of iterations or passes over the training dataset.

(7) The RF model is one of the bagging methods to improve the performance of tree-based regression. A random sample of the predictors is chosen from the full predictors to reduce correlations among trees. There is only one hyper-parameter for the RF model (mtry), which is the number of randomly selected predictors.

(8) The GBM model is a generalization of the tree boosting method to improve prediction performance. Four tuning parameters in the GBM model are n.trees, interaction.depth, shrinkage, and n.minobsinnode. The n.trees is the number of boosting interactions. The interaction.depth is the maximum depth of each tree. The interaction.depth of 1 denotes an additive model and the interaction.depth of 2 denotes a model with 1-way interaction. The shrinkage is also called learning rate or step-size reduction to regularize the regression model. The n.minobsinnode is the minimal number of observations in the terminal nodes of the trees.

(9) The cubist model is a rule-based model using a method similar to boosting learning algorithms [42]. Two parameters need to be tuned: committees and neighbors. The committees are the number of committees (i.e. boosting iterations) to create a sequence of rule-based models in which the current model is affected by the previous model. The neighbors are the number of instances to adjust the rule-based predictions. Note that the determination of committees is not linked to the choice of instances.

(10) The stacking model is one of the ensemble methods to combine several learning models to obtain better results than one single learning model [43]. Firstly, the individual machine learning models (called the first-level models) are created using the available data. Secondly, the predictions from these individual machine learning models are used as inputs to create a combiner algorithm based on the specific methods (called the second-level models), usually simple models, such as linear regression. The first level models are the RF, GBM, and CB, while the second level model is the linear regression to avoid overfitting in this research. The choice of first-level models is dependent on the accuracy and computational cost of learning models.

The concise descriptions of these ten models are presented above in this section. For more information on deep learning models, please refer to [44,45]. For detailed information on other learning models used in this research, please refer to [41,46].

2.2.2. Distributed lag models

The distributed lag (DL) models mean that the energy use of buildings would depend on the current and past values of input variables [47]. These past inputs are also called lagged variables. As a result, the distributed lag models would use the lagged predictors as inputs in creating regression models [48]. The predictors for the distributed lag models with a lag number of 1 include 10 static variables (Table 1), 10 dynamic variables (Table 2), and 9 lagged predictors except for day type (Table 2). Hence, the number of predictors would increase to 29 for the one lag model. Note that the day type would change in the time scale of day and there is no new variable of day type for the lagged models in the hourly time scale. For the 2-hour lag model, extra 9 predictors in the previous two hours (except for day type in Table 2) will be added to the model besides 29 predictors for the 1-hour lag model. The same principle can be applied to the 3-hour and 4-hour lag models. If the lag number is increased to 4, then the number of predictors becomes 56, around 3 times the lag-free models. For learning models, more predictors would require more computational time. Hence, the best three models selected from the lag-free models would be used to create the distributed lag models for both hourly heating and cooling energy of buildings in this case study.

Fig. 3 illustrates the computational procedures for the distributed 3-hour lag models for both heating and cooling energy use of buildings. The building energy at 8:00 on a given day should be firstly predicted since the HVAC system would be turned on at 8:00. The inputs include static parameters and dynamic parameters. The static parameters are 10 factors as listed in Table 1. The dynamic parameters at 8:00 are 10 factors as listed in Table 1. In addition, the dynamic parameters from 5:00 to 7:00 should be also considered due to being a 3-hour lag model. These inputs are entered into the machine learning-based distributed lag model to obtain the energy use at 8:00. The same procedure is also used from 9:00 to 18:00 on a given day. Then the hourly energy use of a building in one year can be predicted from the distributed 3-hour lag models.

2.2.3. Autoregressive distributed lag models

The autoregressive distributed lag (ADL) models [49] would assume that building energy use would depend on both lagged

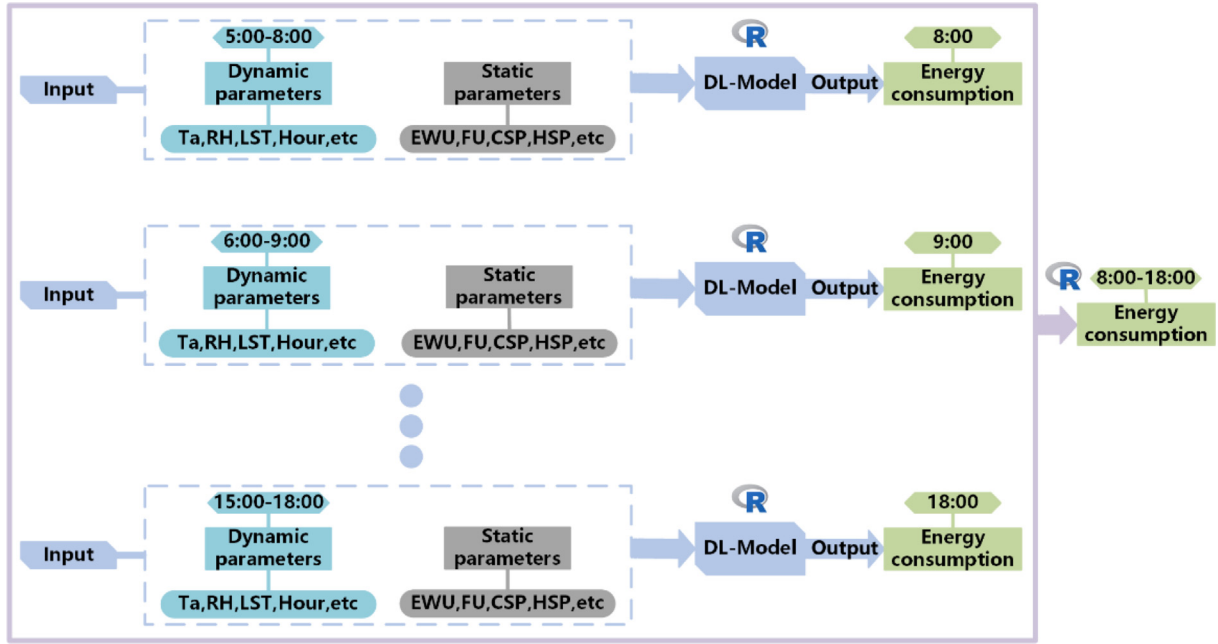


Fig. 3. The distributed 3-hour lag models for time series building energy analysis.

dependent and lagged independent variables. The autoregressive term means the past outputs would also affect the current outputs besides the input variables [47]. Hence, the ADL model would include the lagged dependent and lagged input variables as predictors. The ADL1 refers to the ADL model with one lag. For the ADL1 model, the predictors include 10 static variables (Table 1), 10 dynamic variables (Table 2), 9 lagged predictors except for day type (Table 2), and energy use in the previous hour. As a result, the total number of predictors is 30 for the ADL1 model. The same procedure can be applied to other ADL models. Thus, the ADL4 models have 60 predictors since there are 56 inputs plus 4 energy data among the previous 4 h. Note that the previous-hour energy is required for the ADL model. The determination of previous-hour energy data as predictors would be discussed in the following paragraph.

Fig. 4 illustrates the computational steps for the autoregressive 3-hour lag models applicable to both the heating and cooling energy of buildings. The energy use at 8:00 would be firstly computed since the HVAC system would be switched on at 8:00. There are three types of input data. The first type of data are the static parameters as listed in Table 1. The second type of data are the dynamic parameters in the current hour (8:00) and the previous 3 h (5:00–7:00). The third type of data are the 3-hour previous energy use from 5:00 to 7:00. These energy data would be taken as zero since the HVAC system has been turned off during these three hours. These input data are entered into the autoregressive distributed lag models to obtain the energy use at 8:00. The same procedure can be applied for the prediction of energy use at 9:00. Note that the energy use at 8:00 as one of the inputs is based on the computed value in the last step, but cannot be taken as zero in the case of the computation of energy use at 9:00. This procedure can be used to obtain the hourly energy use from 8:00 to 18:00.

2.3. Data folding

The data folding technique is used to provide a robust estimation of predictive performance from the models selected in the pre-

vious step. The data folding method used in this step is different from the conventional cross-validation [50]. The variable importance would be applied to determine the most important variable affecting building energy use [51]. Then this key variable would be used in the process of data folding analysis to produce a more balanced training set for creating reliable building energy models. The training set would be partitioned into k sets of roughly equal size. The k value would be changed from 1 to 5, which indicates that different sizes of training data sets would be created to assess the stability of predictive performance. The five different sizes are 1/5, 1/4, 1/3, 1/2, and 1 of the whole training set. The corresponding sizes of five data sets are 20, 25, 33, 50, and 100 in this case study. Hence, the change in the predictive performance of the learning models created from different data sizes would be evaluated to provide more robust results. Moreover, this process can be used to choose more accurate learning models from several models selected in the stage of model preselection as described in section 2.2. The process described in this paper has the advantage of being sequential in both two steps. In the step of model pre-selection, several accurate models can be chosen using a small number of data. In the step of data folding, the predictive performance of these models chosen from the first step would be assessed by using the data set with an increase in data size. This process is in line with the features of *meta*-models, which can create the extra training data set by running the engineering-based building energy models more times.

2.4. Performance metrics

Four performance metrics are used to assess the predictive performance of learning models. They are root mean square error (RMSE), coefficient of determination (R^2), mean absolute error (MAE), and root mean square error coefficient of variation (CV (RMSE)), as computed by the following equations.

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{N}} \quad (1)$$

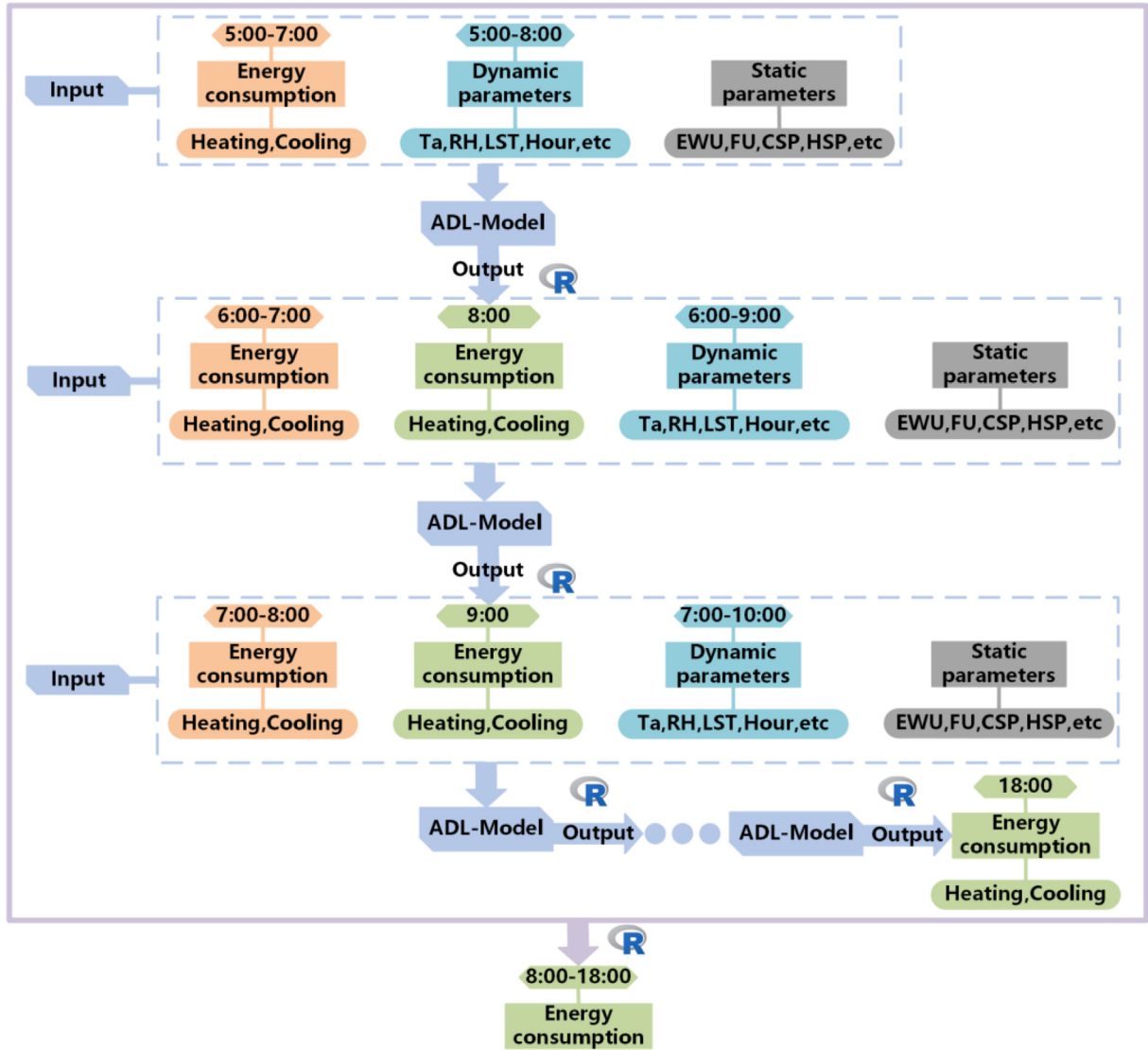


Fig. 4. The autoregressive distribution 3-hour lag models for time series building energy analysis.

$$MAE = \frac{\sum_{i=1}^N |y_i - \hat{y}_i|}{N} \quad (2)$$

$$R^2 = 1 - \sum_{i=1}^N \frac{(y_i - \hat{y}_i)^2}{(y_i - \bar{y}_i)^2} \quad (3)$$

$$CV(RMSE) = \frac{RMSE}{\frac{1}{N} \sum_{i=1}^N y_i} \quad (4)$$

where N represents the total sample size, y_i represents the i^{th} true value, $\bar{y}_i$ represents the average value, and $\hat{y}_i$ represents the i^{th} predicted value.

The RMSE and MAE [52] can be regarded as absolute errors since they have the same unit as the response (kWh in this case study). The MAE is the average of the absolute values of the error between the observation and the prediction values. In contrast, the RMSE is the square root of the average squared errors. As a result, the RMSE is more sensitive to outliers in comparison with the MAE. The R^2 and $CV(RMSE)$ are the relative errors due to being dimensionless. The R^2 is the coefficient of determination to denote

the proportion of the variation in the dependent variable that can be explained by the models. The $CV(RMSE)$ [53] is the normalized RMSE by the average observation values, which is commonly used in the field of building energy simulation.

The computational time is also an important indicator for the *meta*-models in building energy analysis. The computational time used in this study includes both the training and prediction time of learning models. Simulations were performed on a workstation with an Intel Xeon CPU (E5-2650 2.3 GHz) and 32 GB RAM. The parallel computation has been implemented as long as the corresponding R packages support it.

3. Results and discussion

The tuning process of machine learning models would be discussed first. Then the results in the step of model preselection would be presented for heating and cooling energy in sections 3.2 and section 3.3, respectively. Then the predictive performance of selected models would be assessed in the step of data folding. Finally, the practical advice for creating time-series *meta*-models of building energy would be discussed.

3.1. Tuning process of machine learning models

The tuning process can provide useful information on the characteristics of building energy use. This section is focused on the tuning process for the lag-free learning models. Table 3 lists the final tuning parameters from ten lag-free machine learning models. Fig. 5 and Fig. 6 show the tuning process for the GBM and RF models, respectively.

Table 3 lists the optimal tuning parameters from the ten lag-free machine learning models for heating and cooling energy. The meanings of these tuning parameters have been described in section 2.2.1. The differences in these parameters are not significant between heating and cooling energy in this case study. Hence, the model complexity for estimating heating and cooling energy is similar. For the MLR models, there is no parameter to be tuned. For the MARS model, the final tuning parameters are the same for both the heating and cooling energy of the building. For the SVM model, the cost parameter for cooling energy is higher than that for heating energy. As a result, more complex SVM models are required for cooling energy compared to heating energy. For three neural network models, the differences in parameters are small between the heating and cooling energy of buildings. Note that the tuning processes for the DNN and LSTM models are much more difficult and time-consuming in comparison with the conventional ANN. The computational cost of these learning models would be discussed in sections 3.2 and 3.3. For the ensemble models, the parameters are still very similar between heating and cooling energy models. The tuning process for the GBM and RF models would be described in the following paragraph. For the CB models, the final parameters are the same as for the heating and cooling energy models. For stacking models, the linear model is used in the second layer to reduce overfitting.

For the GBM models, the cross-validation RMSE of both heating and cooling energy would be decreased with an increase in boosting iterations as shown in Fig. 5. The RMSE would be also reduced with an increase in interaction depth. Hence, a sufficient number of

trees is necessary to improve model predictive performance. There exist the complex interactions of predictors, which can be confirmed with the degree parameter of MARS models as listed in Table 3. For the RF models, the cross-validation RMSE errors would be decreased with an increase of randomly selected predictors as shown in Fig. 6. After the predictor numbers further increase, the RMSE values would become stable. More predictors are required for the heating energy model compared to the cooling energy model in this case study.

The tuning parameters are discussed for the lag free models. However, it is found that the final tuning parameters in most of the learning models are the same between the lag-free models, the distributed lagged models, and the autoregressive distributed lag models. Hence, the discussion on the other two models would not be presented. More computational cost is required for the two lagged models by considering the lagged effects of both inputs and outputs. Therefore, the ranges of tuning parameters in these two types of models can be reduced based on the tuning results from the lag-free models. This would expedite the computational process of creating reliable *meta*-models for building energy analysis.

3.2. Results of heating energy models from model preselection

This section firstly compares the performance of ten learning models based on 20 data sets from the perspectives of model accuracy and computational cost. Then the best three models would be chosen for the next two subsections to apply the distributed lag models and the autoregressive distributed lag models, respectively.

3.2.1. Results of lag free models

Fig. 7 shows the predictive performance of ten lag-free models for heating energy. The best model is the DNN in terms of four performance metrics, while the linear model performs the worst in this case study. This suggests that there are significant non-linear relationships between inputs and heating energy use for this building. The RMSE from the DNN model has a minimum value of 5 kWh. The next four models (GBM, CB, RF, and Stacking) have similar performance with an RMSE of around 10 kWh, two times the RMSE from the DNN model. The four models (MARS, ANN, LSTM, and SVM) have around the same predictive performance with an RMSE of approximately 20 kWh. In contrast, the RMSE from the linear model is up to 45 kWh. Similar patterns for these ten models can be observed based on the CV(RMSE), R^2 , and MAE as shown in Fig. 7. The CV(RMSE) values from the five best models (DNN, GBM, CB, RF, and stacking) are below 0.15 using the hour time scale, which indicates that these learning models have very good predictive performance. This can be confirmed by the high values of R^2 for these five best models, which is very close to 1.

Table 4 lists the computational time of ten learning models for the cooling energy of buildings. The units of computation time in all tables in this study are seconds (s). As might be expected, the linear model as the simplest model has the least time, while the DNN requires much more computation time with around 400 times compared with the linear model. The second fast model is the MARS, which can account for nonlinear and interaction effects. The next two models (GBM and CB) have a similar computational time of more than 100 s. The conventional neural network model (ANN) is also quite fast in this case study. The next three models (SVM, Stacking, and LSTM) have the same level of computational time around 600 s. The RF model as one of the ensemble models also requires a large amount of time, around 20 min. The learning models are used for the *meta*-models to replace computationally expensive engineering-based models. Hence, computational time is a very important performance indicator, which is different from the conventional time-series energy/load prediction of buildings.

Table 3
The optimal tuning parameters from ten lag-free machine learning models.

Machine learning model	Hyper-parameter	Heating energy	Cooling energy
MARS	Nprune	40	40
	Degree	4	4
SVM	Cost	8	16
	sigma	0.03365	0.03301
ANN	Size	19	15
	Decay	0.00056	0.00024
DNN	Hidden units	c(50,100,150)	c(50,100,150)
	Activation	Tanh	Tanh
LSTM	Layer_lstm	units = 120	units = 100
		activation = tanh	activation = tanh
	Layer_dense	units = 90	units = 80
		activation = tanh	activation = tanh
	Layer_dense	units = 60	units = 50
		activation = elu	activation = elu
	Layer_dense	units = 30	units = 30
		activation = relu	activation = relu
RF	Epochs	260	200
GBM	Mtry	16	13
	N.trees	600	750
	Interaction. depth	12	15
	Shrinkage	0.1	0.1
	N.	10	10
	minobsinnode		
CB	Committees	30	30
	Neighbors	9	9
Stacking	First layer	RF, GBM, CB	RF, GBM, CB
	Second layer	Linear model	Linear model

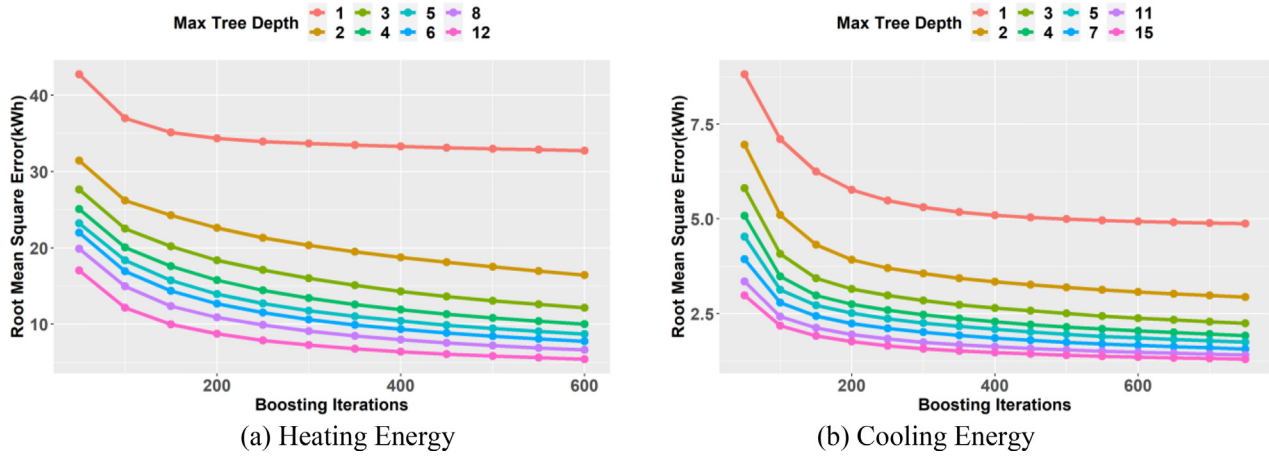


Fig. 5. The tuning process for the GBM (gradient boosting machine) models.

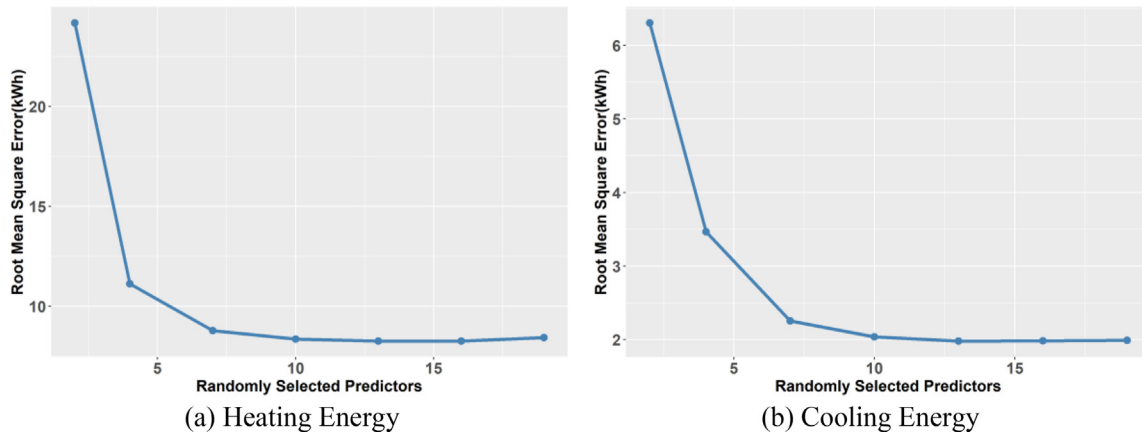


Fig. 6. The tuning process for the RF (random forest) models.

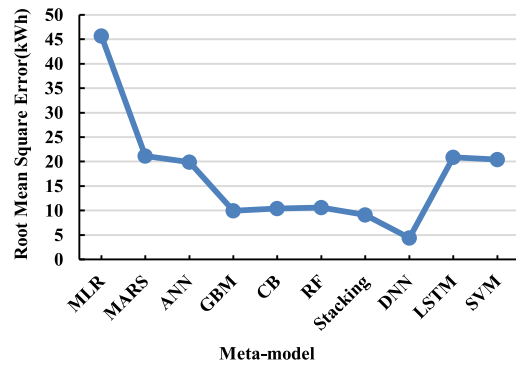
In summary, both the predictive performance and computational cost are two important aspects in assessing the performance of machine learning models. The DNN model performs the best at the expense of the most time-consuming model. The RF model also has good predictive performance with the second time-consuming model. In contrast, the GBM, CB, and stacking models have good predictive capability with a moderate computation time among the ten models. Hence, these three models are chosen for the following two subsections to assess the performance of the time series models for heating energy of buildings.

3.2.2. Results of distributed lag models

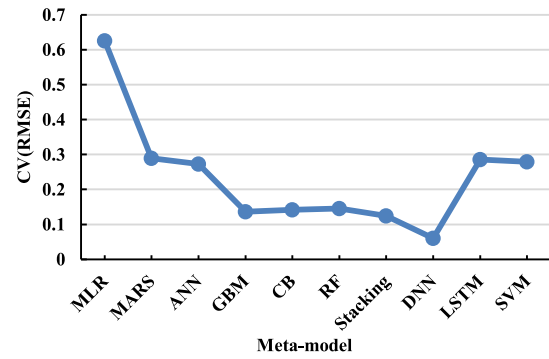
Fig. 8 shows the predictive performance for heating energy from three distributed lag learning models chosen from section 3.2.1 with 1 to 4 lag hours. On the whole, all the models have good performance in terms of CV(RMSE) since all the CV(RMSE) values are less than 0.17. This is also in line with the results in terms of R^2 values, which are above 0.985 from all the learning models. It is interesting to note that the effects of lagged predictors are different for these three models. The GBM-DL0 model refers to the GBM model without the lagged predictors and the GBM-DL1 refers to the GBM model with the 1-hour lagged predictors. This can be applied to other time-series learning models with different lagged hours. For the GBM models, the GBM-DL0 and GBM-DL4 have similar performance, while the other three lagged models have worse predictive performance. As a result, the inclusion of new lagged

predictors does not improve the prediction capability for the GBM models. For the CB and Stacking models, the 1-hour lagged models (CB-DL1 and Stacking-DL1) have the best performance. Hence, it is important to explore the performance of time series models for the various types of learning models. It can also be concluded that the predictive performance would not be enhanced by adding more lagged predictors in more than three previous hours in building energy analysis.

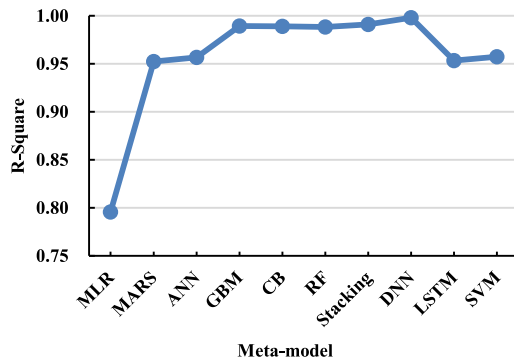
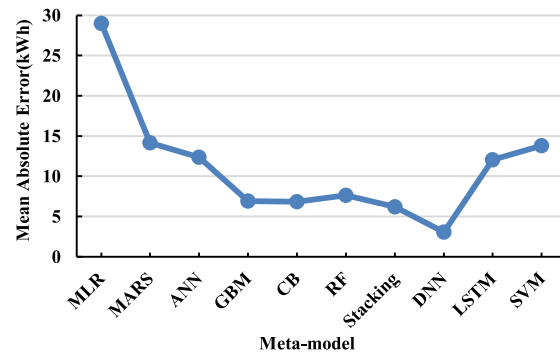
Table 5 shows the computational time for these distributed lag models of heating energy from 1-hour to 4-hour lag. For the GBM model, the computational time with the 2-hour lag model would increase by around 30 % compared with the 1-hour lag model. A similar increase in computational time can be observed for the CB learning models. The change of computational time for the stacking models would be quite irregular because there are several different types of models in the stacking model and the combinational computational time would be hard to estimate. More lagged predictors would lead to a significant increase in computational time. The GBM-DL4 model would require also double the computational time in comparison with the GBM-DL1 model. The computational time would increase to around 22 min for the stacking-DL4 model. In summary, the CB-DL1 and stacking-DL1 models have been chosen as the two best models from these 12 distributed lag models based on the predictive performance and computational cost.



(a) RMSE



(b) CV(RMSE)

(c) R^2 

(d) MAE

Fig. 7. Performance comparison of ten lag-free machine learning models for heating energy.

Table 4

The computational time of ten lag-free heating energy models of buildings.

Meta-model	Time(s)	Meta-model	Time(s)
MLR	8.59	RF	1241.28
MARS	34.55	Stacking	635.59
ANN	318.50	DNN	3729.77
GBM	131.12	LSTM	653.82
CB	113.22	SVM	576.86

3.2.3. Results of autoregressive distributed lag models

The CB models with the autoregressive terms have been found to perform worse than expected due to accumulative errors. Hence, the corresponding stacking model including the CB model does not have good predictive performance. The MARS model due to its good predictive performance and fast computation has been used to replace the CB model in the new stacking model. Table 6 lists the performance and computational time for heating energy from MARS-ADL1 and CB-ADL1 models by considering both the autoregressive terms and lagged predictors. The MARS-DL1 performs much better than the CB-ADL1 in terms of both the predictive capability and computational time. The CB models are not used in the section and there are only two models (GBM and new stacking) as illustrated in Fig. 9.

Fig. 9 shows the predictive performance of autoregressive distributed lag models with 1 to 4 lags for heating energy based on the GBM and stacking models. The predictive performance would not be improved after adding the autoregressive terms and lagged predictors in terms of RMSE. For the GBM model, the GBM-ADL0 model performs the best. This is also the case for the stacking

model. Similar trends can be observed for the other three performance metrics (CV(RMSE), R^2 , and MAE). This is due to the accumulative errors for the autoregressive models. The errors of predictive heating energy in the previous hours would be added in the current and future hours.

The computational time of autoregressive distributed lag models for predicting heating energy is listed in Table 7. For the GBM model, the computational time for the GBM-ADL2 increases by around 36 % compared to the GBM-ADL1. In contrast, the computation time for the stacking-ADL2 model increases by only 7 % in comparison with the stacking-ADL1. Note that the computational time in Table 7 is not comparable with the computational time in Table 5. This is because the stacking models include different learning models as described at the beginning of this subsection.

For the stacking models, the stacking-ADL1 has the best predictive performance with the least computational time. For the GBM model, the GBM-ADL1 model has a slight reduction in prediction performance in comparison with the GBM-ADL2, while the computational time for the GBM-ADL1 is significantly less than the time for the GBM-ADL2. Hence, the stacking-ADL1 and the GBM-ADL1 models have been selected for the final comparison of the model performance in the following sections.

3.2.4. Comparison of model performance for heating energy

Table 8 lists the predictive performance and computational time for heating energy of seven meta-models selected from the previous three subsections. These models include three models from the lag-free model, two distributed lag models, and two autoregressive distributed lag models. The best three models are stacking-DL1, stacking, and CB-DL1 models from the perspective

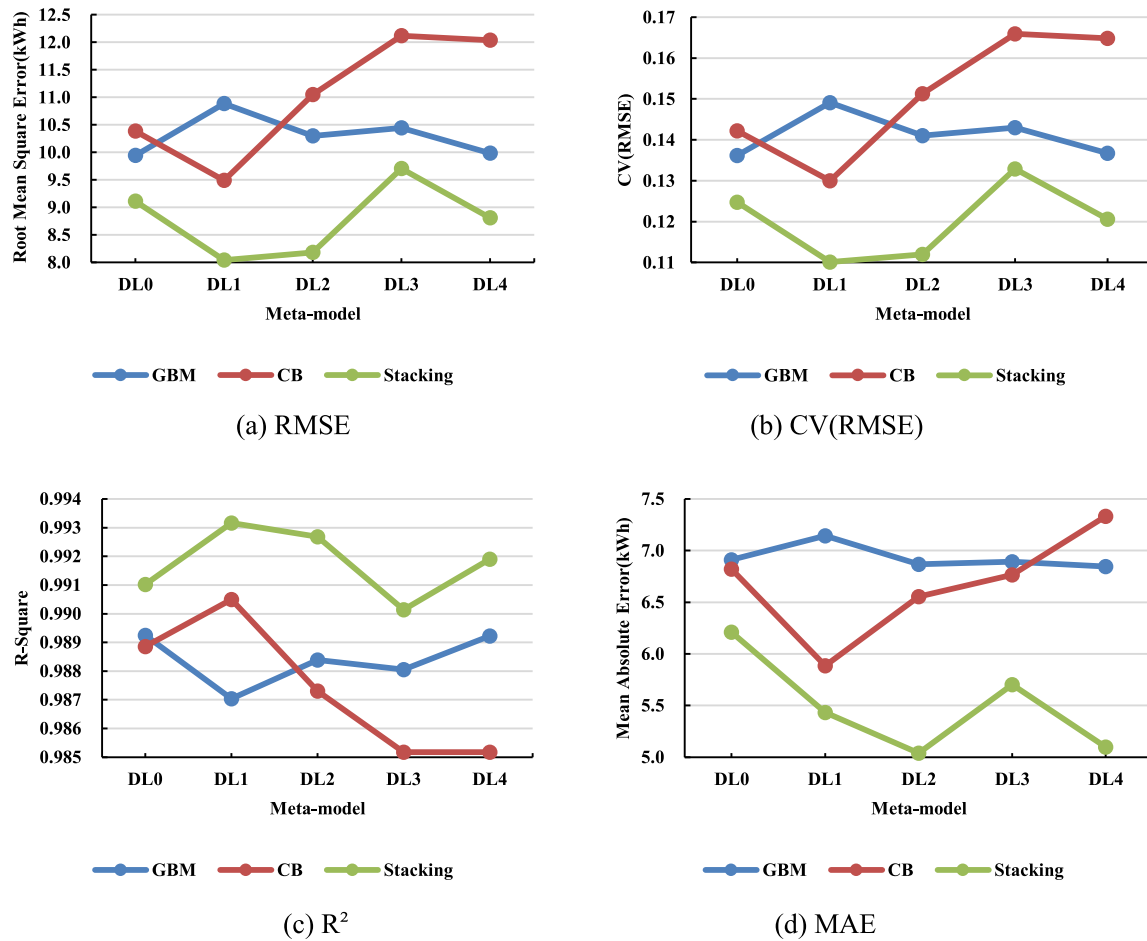


Fig. 8. Performance comparison of heating energy models with distribution lag of 0–4 h (DL0, lag-free model; DLn, n-hour lag model).

Table 5

The computational time of creating machine learning models for heating energy with the distribution 1–4 h lags.

Meta-model	Time(s)	Meta-model	Time(s)	Meta-model	Time(s)
GBM	DL1 196.69	CB	DL1 162.83	Stacking	DL1 720.79
	DL2 257.44		DL2 215.73		DL2 783.08
	DL3 320.55		DL3 254.01		DL3 971.89
	DL4 386.49		DL4 284.65		DL4 1290.66

Table 6

Predictive performance and computational time for the MARS-ADL1 and CB-ADL1 models for heating energy analysis.

Meta-model	RMSE	CV(RMSE)	R ²	MAE	Time(s)
MARS-ADL1	19.480	0.267	0.958	13.494	56.020
CB-ADL1	67.409	0.923	0.605	31.152	663.230

of predictive performance. The R^2 values from all three models are above 0.99, which indicates very good prediction capability. The computational time for the CB-DL1 is significantly less than (around three-third) that for the stacking and stacking-DL1 model. Hence, the CB-DL1 model has been selected as one of the best models in terms of both prediction capability and computational efficiency. In terms of computational time, the GBM and CB models are very similar, at approximately 2 min. The GBM model has slightly better predictions in comparison with the CB model. Hence, the GBM model is also selected as one of the best models for data folding analysis. Therefore, the CB-DL1 and GBM models are two models selected to assess their performance in the larger dataset as will be discussed in section 3.4.

3.3. Results of cooling energy models from model preselection

This section firstly compares the performance of ten learning models for cooling energy based on 20 data sets from the perspectives of model accuracy and computational cost. Then the best three models for cooling energy would be chosen for the next two subsections to apply the distributed lag models and the autoregressive distributed lag models, respectively.

3.3.1. Results of lag free models

Fig. 10 shows the predictive performance of ten meta-models for cooling energy of buildings. The two best models in terms of predictive performance are the DNN and LSTM, which have the

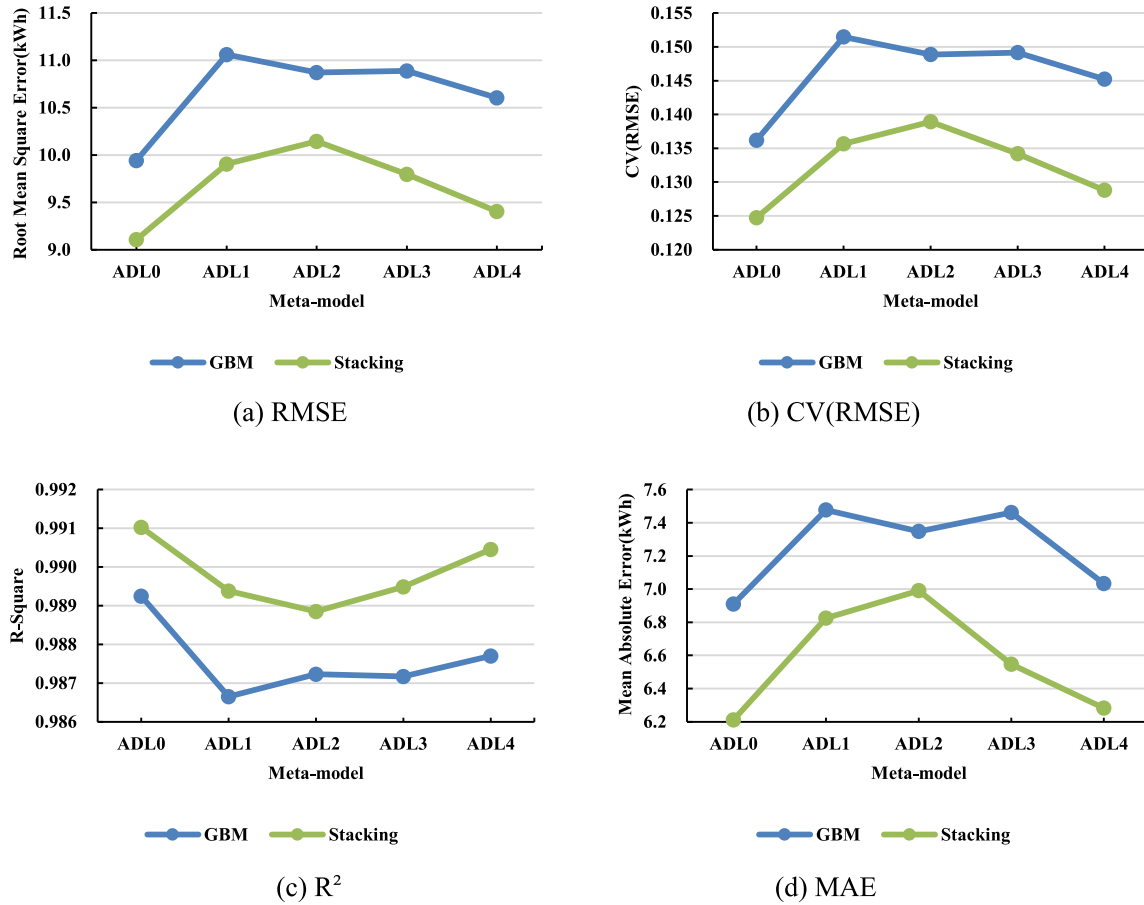


Fig. 9. Performance comparison of heating energy models with autoregressive distributed lag models (ADL0, without lag; ADLn, *n*-hour lag and autoregressive models).

Table 7

The computational time of autoregressive distributed lag models for heating energy based on the GBM and stacking learning models.

Meta-model			Meta-model		
Time(s)			Time(s)		
GBM	ADL1	205.34	Stacking-new	ADL1	788.19
	ADL2	279.86		ADL2	849.03
	ADL3	342.90		ADL3	919.65
	ADL4	434.91		ADL4	977.08

Table 8

Predictive performance and computational time for the top seven models for time series heating energy analysis.

Meta-model	RMSE	CV(RMSE)	R ²	MAE	Time(s)
GBM	9.943	0.136	0.989	6.910	131.120
CB	10.383	0.142	0.989	6.820	113.22
Stacking	9.108	0.125	0.991	6.211	635.590
CB-DL1	9.488	0.130	0.991	5.887	162.830
Stacking-DL1	8.041	0.110	0.993	5.434	720.790
GBM-ADL1	11.059	0.151	0.987	7.477	205.340
Stacking-ADL1	9.905	0.136	0.989	6.824	788.190

CV(RMSE) values of 0.046 and 0.051, respectively. The R² values for the DNN and LSTM are 0.978 and 0.970, respectively, which indicates a very good prediction performance. The next three models are stacking, CB, and GBM, which have R² values above 0.9. The CV(RMSE) values for these three models are around 0.10. The following four models are RF, MLR, MARS, and SVM, in terms of prediction capability. The CV(RMSE) and R² for the MLR model are 0.11 and 0.84, which suggests that the linear model has moderate performance. The ANN model performs the worst in this case study due to overfitting.

Table 9 lists the computational time of ten learning models for the cooling energy of buildings. The fastest model is the MLR with around 8 s. The next three models (MARS, CB, and GBM) have a similar computational time, slightly over 1 min. The computation time for ANN, LSTM, and stacking models is less than 9 min. The following two models (SVM and RF) require over 13 min. The DNN is the most consuming model with around 1 h.

Predictive performance and computational time are considered to choosing the models for further analysis as will be described in sections 3.3.2 and 3.3.3. The LSTM model has been selected as one

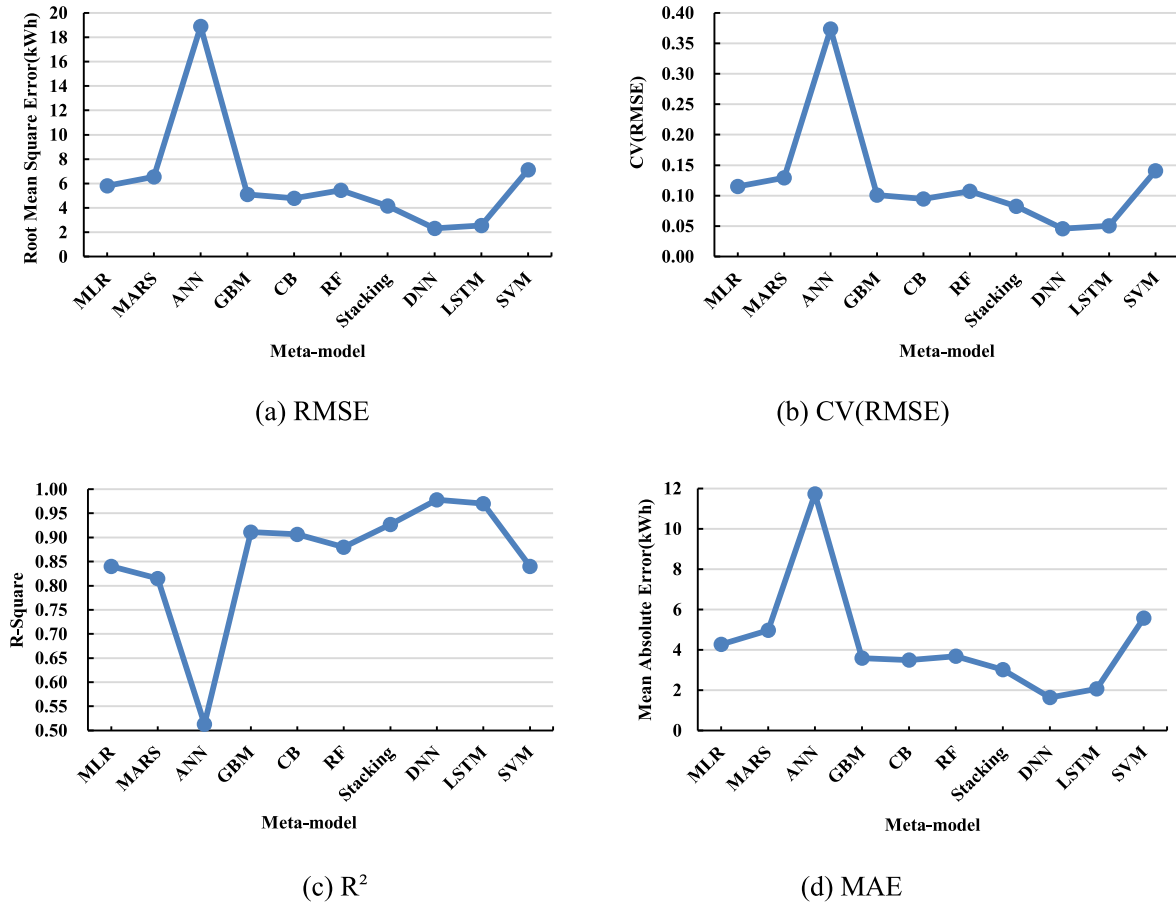


Fig. 10. Performance comparison of ten lag-free machine learning models for cooling energy.

Table 9

The computational time of ten lag-free cooling energy models of buildings.

Meta-model	Time(s)	Meta-model	Time(s)
MLR	8.19	RF	928.30
MARS	63.97	Stacking	501.06
ANN	407.01	DNN	3071.31
GBM	83.77	LSTM	433.70
CB	76.99	SVM	797.42

of the best models with good predictive performance and moderate computational cost. The DNN model is not chosen due to the long computational time although it also has a very good prediction performance. The GBM and CB models are also selected since they have both good predictive capability and fast computation. Hence, the LSTM, GBM, and CB models are used to construct the distributed lag and autoregressive distributed lag time series models for building cooling energy.

3.3.2. Results of distributed lag models

The performance of LSTM, GBM, and CB models with distributed lags would be explored in this section. It is found that the predictive performance of LSTM models by adding lagged predictors would be worse than the lag-free LSTM models. The RMSE of the LSTM model would be increased from 2.56 to 4.90 by adding the 1-hour lag predictors. The predictive performance for the LSTM-DL2 model would be further decreased and the computational time would be almost double compared with the lag free LSTM model. As a result, the performance of LSTM models would not be studied in this subsection. Fig. 11 shows the predictive performance for

cooling energy from GBM and CB models with the distributed lag predictors.

The distributed models with 1-hour lag predictors would perform better than the other *meta*-models for the cooling energy of buildings. This conclusion applies to both GBM and CB models. The RMSE of GBM models would decrease by around 15 % from the GBM-DL0 to the GBM-DL1. The RMSE would increase after adding more lagged predictors for the GBM models. This is in line with the other three performance metrics, including CV(RMSE), R^2 , and MAE. This trend can be observed in the CB models. The R^2 for the CB-DL1 is over 0.93, which is the maximum value among the distributed lag CB models. Hence, the predictive performance of GBM and CB models can be improved by adding 1-hour lag predictors. More lagged predictors in previous hours would reduce the prediction capability of *meta*-models for cooling energy.

Table 10 lists the computational time for the distributed lag GBM and CB models for cooling energy. The computational time for the GBM-DL2 model would increase by around 9 % compared to the GBM-DL1 due to the increase in the number of predictors. The GBM-DL4 requires around double the computational time of the GBM-DL1. In contrast, the CB-DL4 model would increase by around 44 % compared to the CB-DL1. The best models for the distributed lag cooling energy models would be the GBM-DL1 and CB-DL1 models since these two models have the best prediction performance with the least computational time from 1-hour to 4-hour lag models.

3.3.3. Results of autoregressive distributed lag models

This section focuses on the model performance of autoregressive distributed lag models for cooling energy. The LSTM model

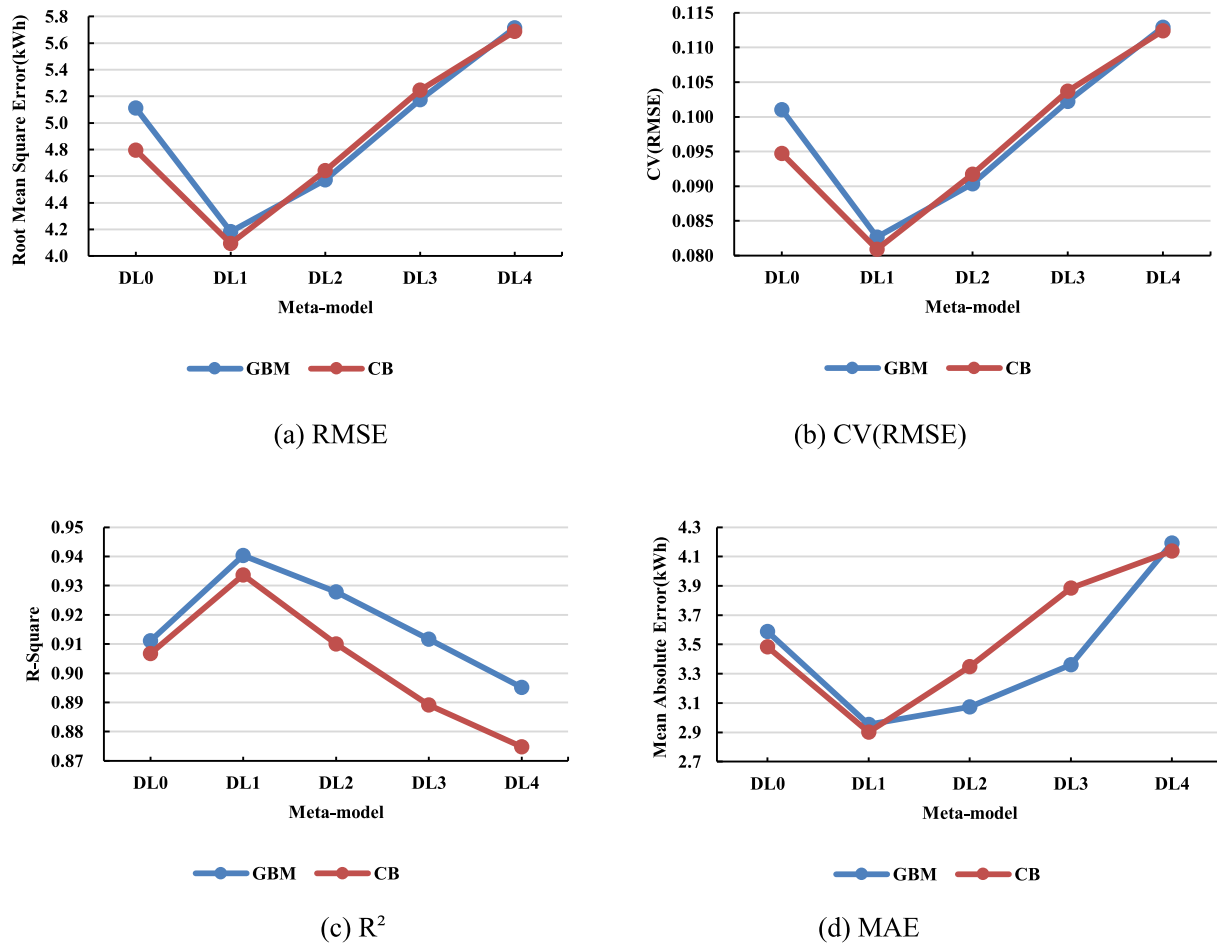


Fig. 11. Performance comparison of cooling energy models with distribution lag of 0–4 h (DL0, lag-free model; DLn, n-hour lag model).

has not been used here for the same reason as discussed in section 3.3.2. Fig. 12 shows the predictive performance of GBM and CB autoregressive distributed lag models for cooling energy. For the GBM and CB models, the change in predictive performance with an increase in time lag is different. For the CB model, the CB-ADL1 with 1-hour lag has the best performance among the four ADL models. The increase of more time lags would lead to a decrease in prediction performance for building cooling energy. In contrast, the predictive performance of ADL models would vary slightly from GBM-ADL0 to GBM-ADL3. When the time lag increases to 4 h, the GBM-ADL4 performs poorly in this case study. Hence, the inclusion of previous energy data and input variables in the previous hours does not result in a significant increase in predictive performance for the GBM models.

The CB-ADL1 model has the best predictive performance with moderate computational time among the 8 ADL models. Hence, the CB-ADL1 has been selected as one of the best models. The least computational time is the GBM-ADL1 although its prediction performance with the R^2 over 0.91 is acceptable. Thence, the CB-ADL1 and GBM-ADL1 models have been chosen two models for further analysis as will be described in section 3.3.4.

Table 10 lists the computational time of cooling energy analysis for the autoregressive distributed lag models. For both the GBM and CB models, the computational time would increase with an increase in time lags. The patterns of computational time for these two models are very different. For the GBM models, the GBM-ADL4 model requires two times the computational cost in comparison with the GBM-ADL1 model. The computational time

for the CB-ADL4 would increase by around 35 % compared to the CB-ADL1 model. The patterns of change in the computational time between the DL and ADL models are very different in these two models. For the GBM models, the increase in computational time is not significant from the DL to ADL models. In contrast, the CB models have a significant increase in computational time from the DL to ADL models. The computational time for the CB-ADL1 is around 4 times of CB-DL1. The computational time includes the construction model time and prediction time. The number of data set is the same for the CB-DL1 and CB-ADL1 models. The increase in the number of inputs is only one from the CB-DL1 to ADL-1. Hence, the time for the model construction is almost the same. Then the increase in computational time is mainly due to the prediction energy using the CB models. Note that parallel computing does not help reduce the computational cost for the ADL models. This is because the energy prediction in the current hour depends on the energy data in the previous hour. Hence, the sequential features make it hard to further expedite the computational time in predicting the energy use when using the ADL models.

3.3.4. Comparison of model performance for cooling energy

Table 11 lists the predictive performance and the computational cost for the best seven models of cooling energy, including three lag free models, two DL models, and two ADL models. The two best models are CB-ADL1 and LSTM in terms of prediction performance with R^2 of around 0.97 and CV(RMSE) of around 0.05. The computational time of LSTM is less than around 40 % of CB-ADL1.

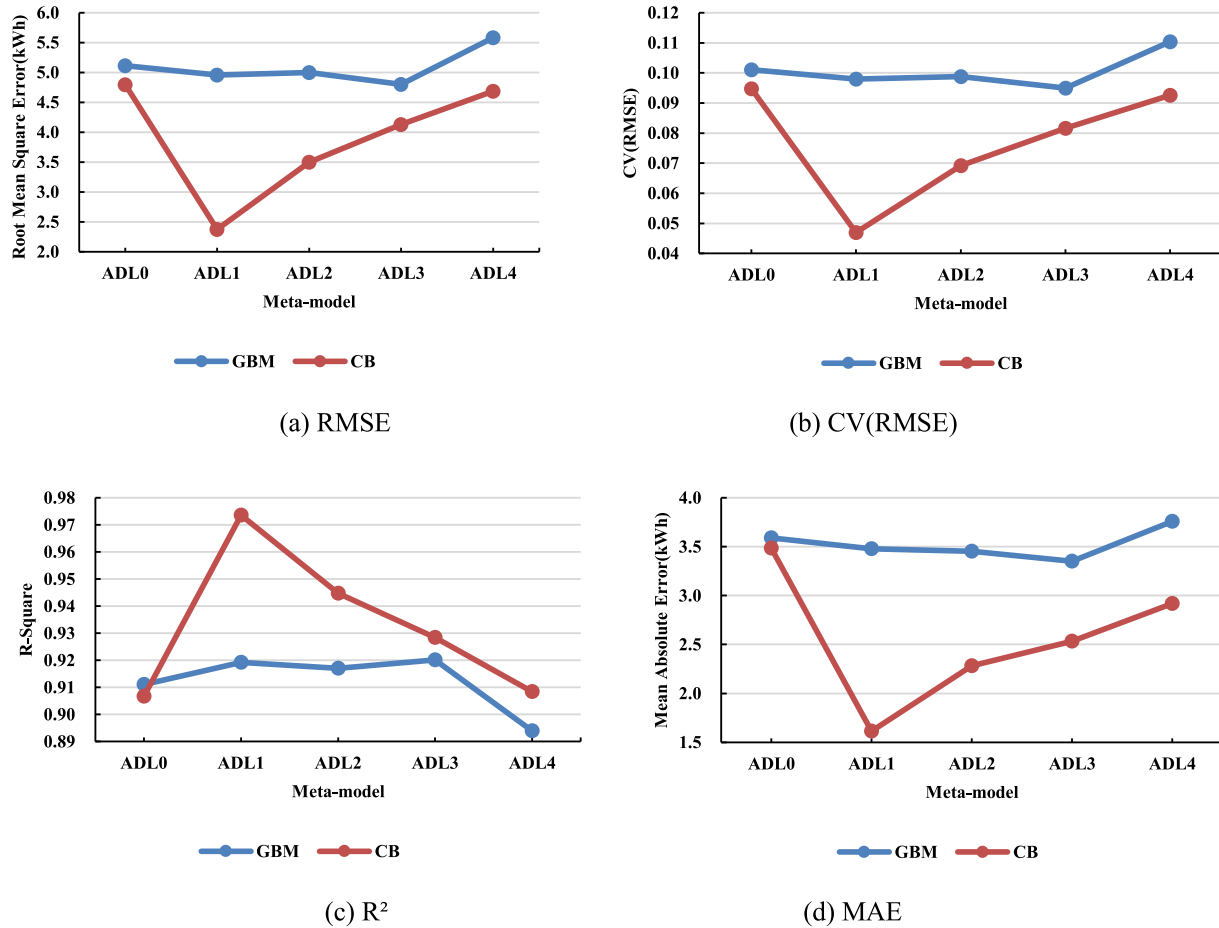


Fig. 12. Performance comparison of cooling energy models with autoregressive distributed lag models (ADL0, without lag; ADL1, n -hour lag and autoregressive models).

Table 10

The computational time of cooling energy models with distribution lag from 1 h to 4 h for the DL (distributed lag) and ADL (autoregressive distributed lag) models.

Meta-model	DL model	Time(s)	ADL model	Time(s)
GBM	DL1	130.22	ADL1	144.03
	DL2	142.20	ADL2	192.99
	DL3	221.36	ADL3	234.15
	DL4	267.49	ADL4	294.18
CB	DL1	153.65	ADL1	604.25
	DL2	174.07	ADL2	687.36
	DL3	198.00	ADL3	758.27
	DL4	220.75	ADL4	818.54

Table 11

Predictive performance and computational time for the top seven models for time series cooling energy analysis.

Meta-model	RMSE	CV(RMSE)	R ²	MAE	Time(s)
GBM	5.113	0.101	0.911	3.589	83.770
CB	4.794	0.095	0.907	3.484	76.990
LSTM	2.565	0.051	0.970	2.064	433.700
GBM-DL1	4.183	0.083	0.940	2.954	130.220
CB-DL1	4.094	0.081	0.934	2.901	153.650
GBM-ADL1	4.958	0.098	0.919	3.478	144.030
CB-ADL1	2.375	0.047	0.974	1.615	604.250

As a result, the LSTM has been chosen one of the best models for cooling energy due to its high prediction capability and moderate computational cost. The next two models are CB-DL1 and GBM-DL1 based on the prediction capability with the R^2 around 0.94.

These two models also have fast computation between 2 and 3 min. The GBM-DL1 model has been chosen for data folding analysis due to its good prediction performance and fast computational time.

3.4. Results from data folding analysis

This section would assess the predictive performance of learning models selected from sections 3.2 and 3.3 by using the data folding technique.

3.4.1. Heating energy models

The variable importance is conducted to determine the most important variables affecting the heating energy of buildings. Fig. 13 illustrates the results of variable importance using the CB-DL1 model. For the hourly dynamic variable, the suffix L0 after the underscore denotes the value in the current hour, while the suffix L1 denotes the value in the previous lag hour. For example, the Ta_L0 is the dry-bulb temperature in the current hour, whereas the Ta_L1 is the dry-bulb temperature in the previous hour. The DT (day type) is the most important factor influencing building heating energy. The same conclusion can be obtained from the GBM model. The following four variables are Ta_L1, DFS_L1, LST_L0, and Ta_L0, which indicates that the dry-bulb temperature and lighting switch time in the previous hours also has important influences on the heating energy of buildings. The aim of using variable importance analysis is to choose the most important factor to choose parts of data for performance evaluation. Hence, more detailed descriptions of the results of importance ranking would not be discussed in this section.

Table 12 lists the predictive performance and computational time of GBM and CB-DL1 models for heating energy with different data sizes. Fig. 14 illustrates the difference in hourly heating energy between the EnergyPlus simulation and learning models. The differences between engineering-based simulation and GBM learning models are very small when using the full data set as shown in Fig. 14 (a). When only 1/5 data set is only used, the energy data from the GBM models are still close to the simulation model as illustrated in Fig. 14 (b). For the GBM model, the increase in data size would not improve the predictive performance of learning models, although the computational cost has increased steadily. The R^2 values of using the full and 1/5 data set are the same, while the computational time would increase around 6.5 times. Then it is unnecessary to create more training data sets for the GBM model. In contrast, the predictive performance for the CB-DL1 models would slightly increase with an increase in data size. The CV(RMSE) of CB-DL models created from the 1/5 to the full training data set would be reduced by around 33 % at the expense of increased computational cost by around three times. The CV(RMSE) values for the CB-DL1 models are less than 0.10 and the R^2 values are above 0.995, which suggests there is a high predictive capability for hourly models of building energy. This can be confirmed in Fig. 14 (c) and Fig. 14 (d). The differences between simulation models and CB-DL1 learning models are very small for full and 1/5 data sets. The construction of the training

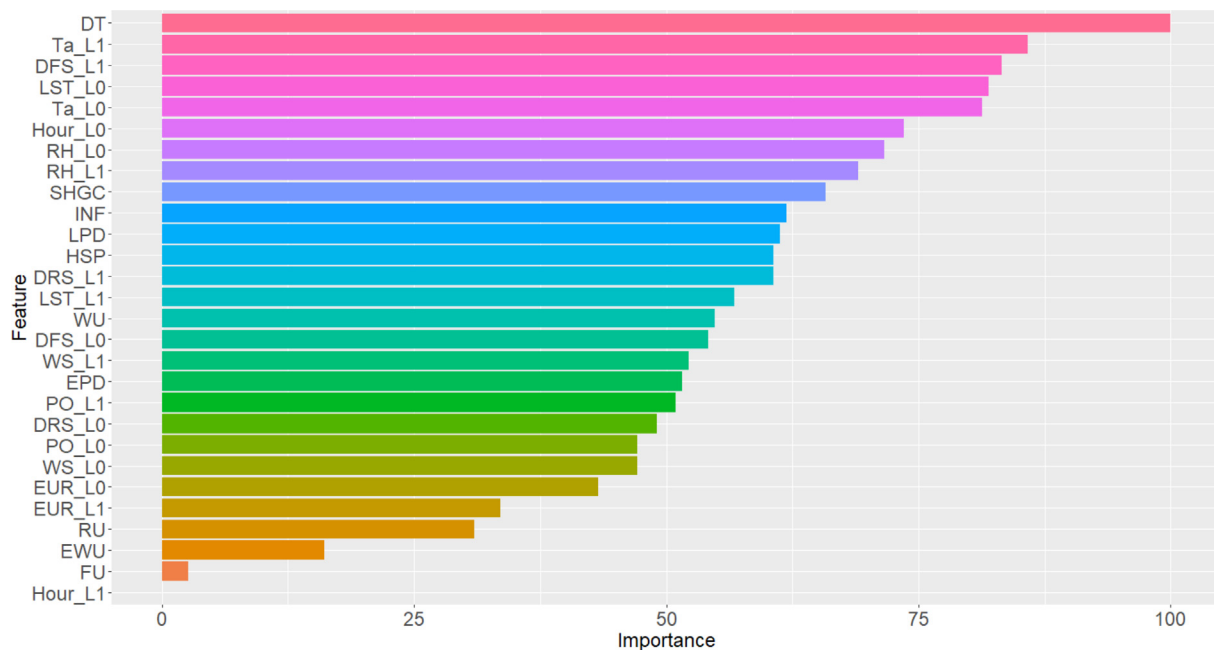
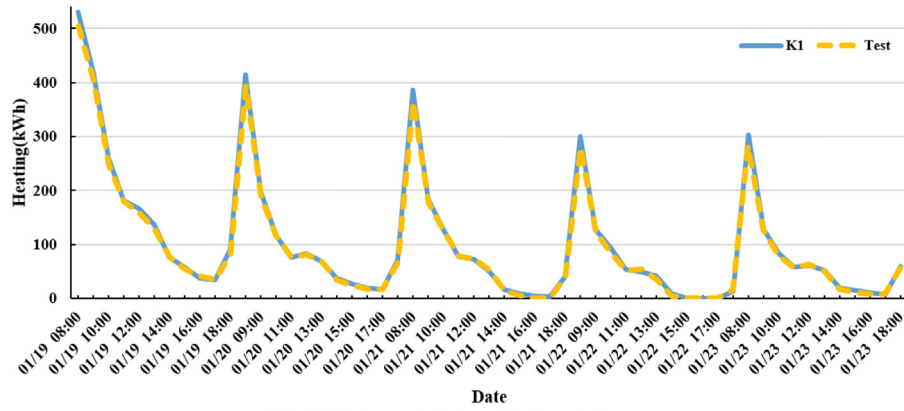


Fig. 13. Variable importance for heating energy using the CB-DL1 (1-hour lag) model.

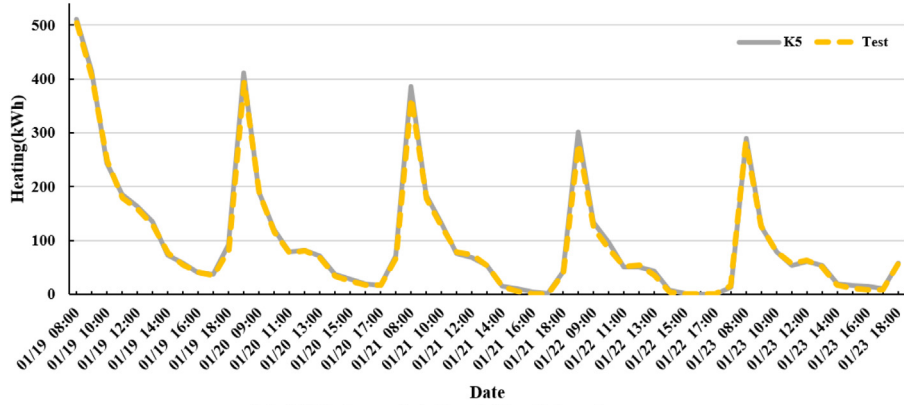
Table 12

Performance indicators and computational time of heating energy consumption models with different data sizes.

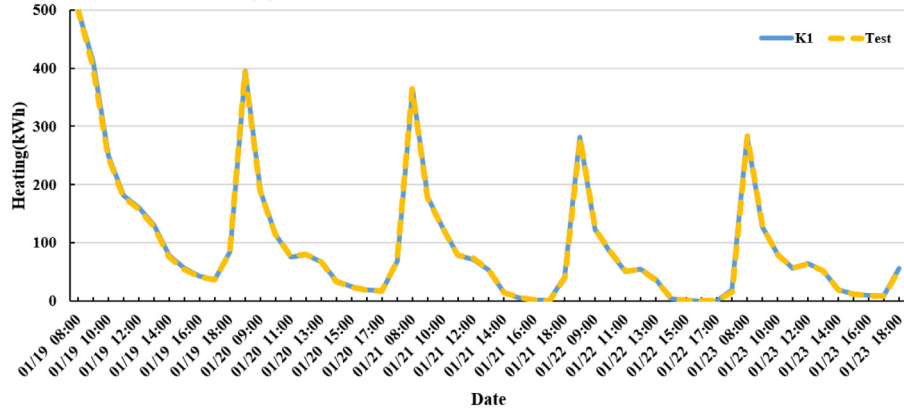
Meta-model	Data-size	RMSE	CV(RMSE)	R^2	MAE	Time(s)
GBM	1	8.031	0.117	0.993	5.459	1158.920
	1/2	8.174	0.119	0.993	5.487	441.210
	1/3	8.029	0.117	0.993	5.460	269.330
	1/4	7.862	0.115	0.994	5.307	204.880
	1/5	8.198	0.119	0.993	5.512	154.780
CB-DL1	1	3.442	0.050	0.999	2.003	1166.730
	1/2	4.500	0.066	0.998	2.550	766.010
	1/3	4.587	0.067	0.998	2.709	561.360
	1/4	4.898	0.071	0.997	2.741	443.860
	1/5	5.160	0.075	0.997	2.943	380.670



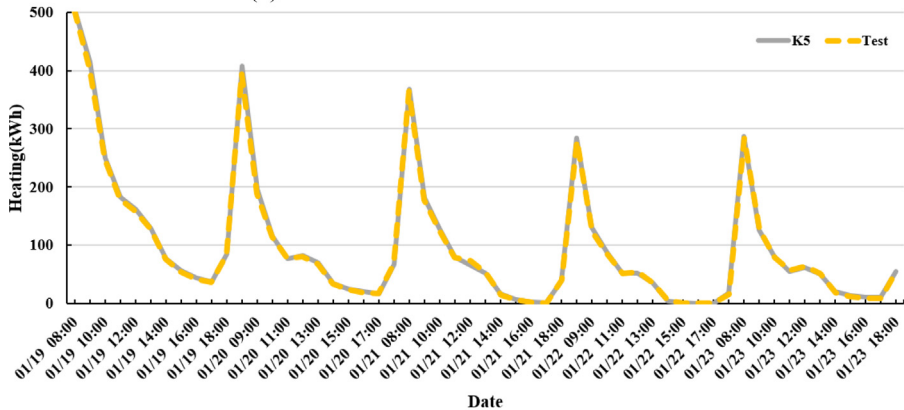
(a) GBM model for all the data set



(b) GBM model for 1/5 of the data set



(c) CB-DL1 model for all the data set



(d) CB-DL1 model for 1/5 of the data set

Fig. 14. Prediction from machine learning models and simulated values from engineering-based models for heating energy.

set can be stopped as long as the accuracy of the learning models has met the project requirements.

It would be interesting to compare the performance of models obtained from this research with the conventional benchmark time series model analysis since there are marked periodic fluctuations at a daily scale as shown in Fig. 14. The previous-day data model is used as a benchmark in this study. The CV(RMSE) and R^2 values from this previous-day model are 0.698 and 0.735, respectively, which are much worse than the models listed in Table 12. Hence, the models created from this study have much better performance in comparison with the conventional time series benchmarks.

3.4.2. Cooling energy models

Fig. 15 shows the results of variable importance for building cooling energy using the GBM-DL1 model. Ta_L0 (dry bulb temperature in the current hour) is the dominant variable influencing the hourly cooling energy of buildings. The next important variable is the SGHC of windows. These first two key variables are much more important compared with the remaining variables in predicting the hourly cooling energy of buildings. This trend is different from the results of variable importance for heating energy as shown in Fig. 13 in which at least ten variables have important influences on the heating energy. As a result, the Ta_L0 is used to partition the data set to obtain more balanced data for creating learning models.

Table 13 shows the predictive performance and computational time for cooling energy using the different data sizes. Fig. 16 illustrates the comparison of cooling energy in five days between engineering-based EnergyPlus simulation and learning model prediction. For the LSTM models, the predictive performance would gradually increase with an increase in sample size although there are fluctuations in performance indicators. The CV(RMSE) of LSTM models would decrease from 0.043 to 0.016 when the sample size changes from the 1/5 to the full data set. The corresponding R^2 would increase from 0.971 to 0.996. The differences in cooling energy between the EnergyPlus simulation and LSTM models for the full data set are very small as shown in Fig. 16 (a). The differences become slightly larger when the data size decreases to the 1/5 data set as illustrated in Fig. 16 (b). The computational cost would increase around four times when the data size increases from 1/5 to the full data set. The choice of the final model for cooling energy should be balanced between model accuracy and computational cost.

For the GBM-DL1 models, the predictive performance would change slightly when the data size increases. The CV(RMSE) would vary between 0.035 and 0.038 when the data size changes from 1/5 to the full data set. The R^2 also fluctuates around 0.98. This is in agreement with the visual comparison as shown in Fig. 16 (c) and Fig. 16 (d). The differences in hourly cooling energy between the EnergyPlus simulation and GBM-DL models are very similar

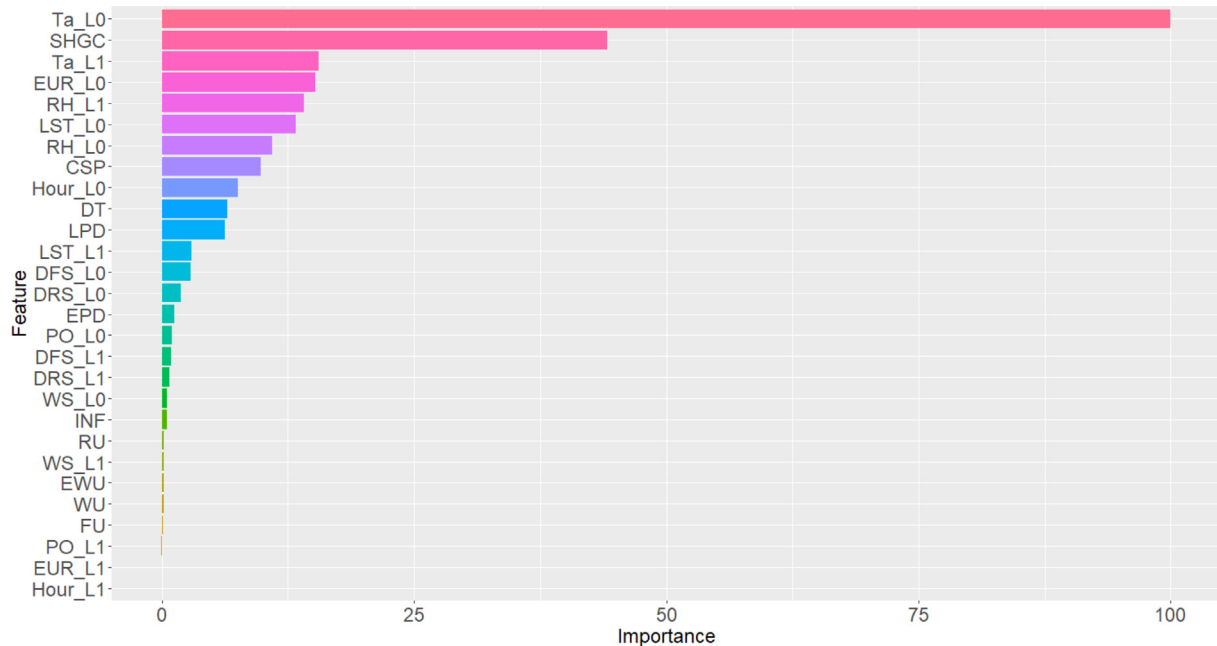
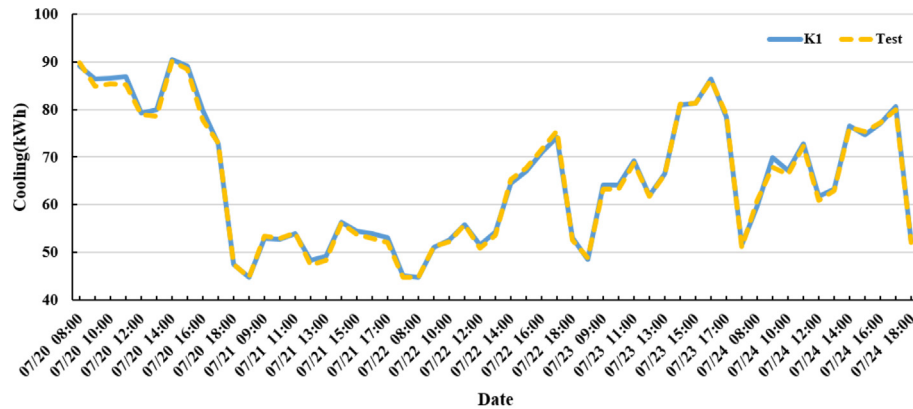


Fig. 15. Variable importance for cooling energy using the GBM-DL1 (1-hour lag) model.

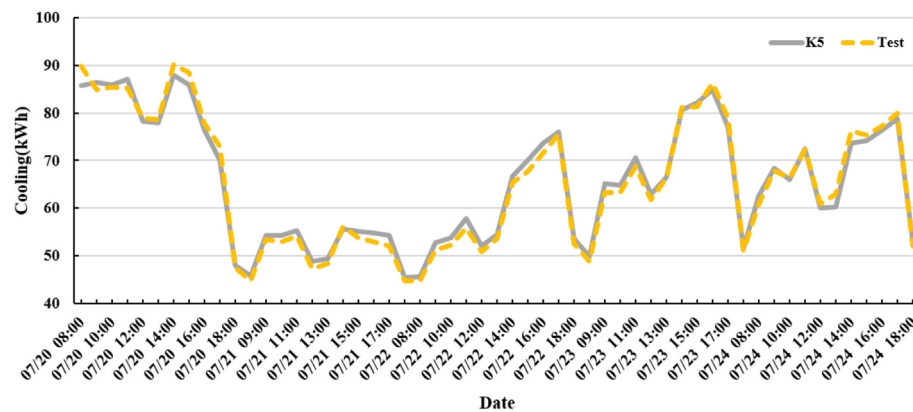
Table 13

Performance indicators and computational time of cooling energy consumption models with different data sizes.

Meta-model	Data-size	RMSE	CV(RMSE)	R^2	MAE	Time(s)
LSTM	1	0.926	0.016	0.996	0.583	2445.470
	1/2	0.896	0.016	0.996	0.568	1286.530
	1/3	1.221	0.022	0.993	0.823	946.480
	1/4	1.176	0.021	0.993	0.757	760.160
	1/5	2.403	0.043	0.971	1.650	604.600
GBM-DL1	1	1.981	0.035	0.980	1.502	1604.430
	1/2	2.036	0.036	0.979	1.513	558.340
	1/3	2.056	0.037	0.979	1.522	348.660
	1/4	1.979	0.035	0.980	1.457	262.440
	1/5	2.131	0.038	0.977	1.574	202.620



(a) LSTM model for the full data set



(b) LSTM model for the 1/5 data set

Fig. 16. Prediction from machine learning models and simulated values from engineering-based models for cooling energy.

in these two subfigures using different data sizes. In contrast, the computational cost would increase around eight times when the data size increase from 1/5 to the full data set. Hence, it would be unnecessary to construct more data sets for improving model accuracy for the GBM-DL1 models.

The simple previous-day model is commonly used in time series analysis as the benchmark to assess the predictive performance of complex time series models. The CV(RMSE) and R^2 values from the previous-day cooling energy model are 0.206 and 0.545, respectively. Hence, all the models for cooling energy listed in Table 13 have better predictive capability compared to the previous-day model. It would be interesting to compare the previous-day for heating (section 3.4.1) and cooling. The performance from the previous-day cooling energy model is worse than that from the heating energy model. This suggests that the patterns for cooling energy would be much more complex compared to heating energy. As discussed in section 3.4.1 and this section, it is necessary to create complex models by considering input influences to obtain good performance models in building energy models in comparison with the simple benchmark time series models.

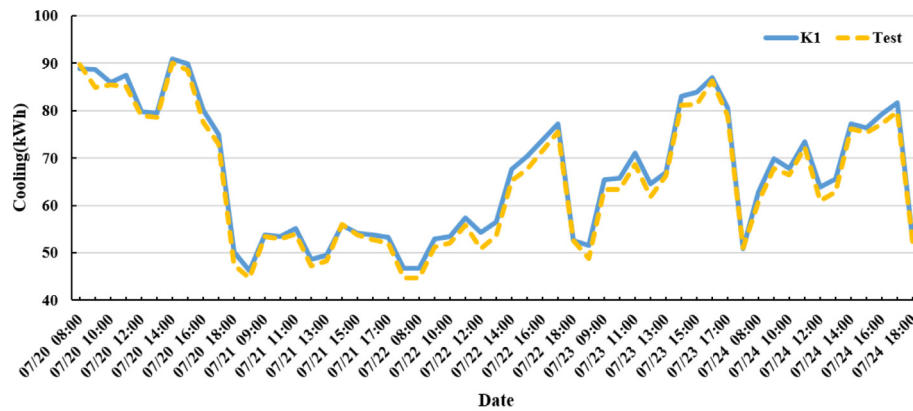
3.5. Practical advice in the construction of building energy time series meta-models

This section discusses the practical advice for creating machine learning-based time series *meta-models* in building energy analysis. The topics include the generality of the method proposed, the choice of time series models, the difference between conventional data-driven models and *meta-models*, the use of sequential features in *meta-model* construction, the choice of learning models, and the application of hourly *meta-models*.

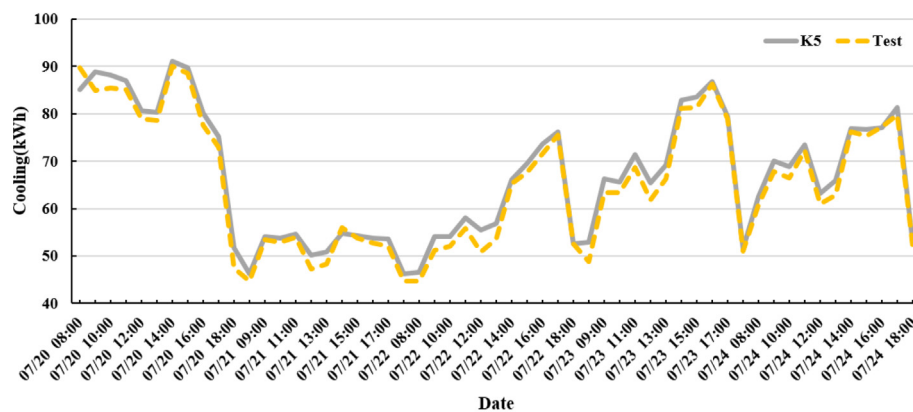
The method illustrated in Fig. 1 can be applied to the different buildings in various locations due to its good generality although this method is demonstrated by a case study as described in section 2.1. The generality of the method mainly includes the choice of learning models and the selection of time features of time series models. The choice of lag-free methods can be easily expanded by adding more new and advanced machine learning algorithms to increase the predictive capability of models except for ten learning models used in section 2.2.1. The different time features of energy use in buildings are considered, including lag-free, distributed lag, and autoregressive distributed lag models. Hence, the models created here can be more suitable for predicting energy use in different buildings.

It is necessary to explore the characteristics of (autoregressive) distributed lag models in creating time series building energy models. The predictive performance of lag-free models should be first evaluated since they are fast run due to the fewer input numbers in comparison with the (autoregressive) distributed lag models. The distributed lag models would perform better compared to the autoregressive distributed lag models due to the accumulative errors of autoregressive models. If the project is limited by the time and computational cost, the lag free and distributed lag machine learning models can be applied in constructing building energy models. The lag number can be also explored using the cross-validation technique. The higher lag number over four is unlikely to be useful to improve the predictive performance in creating *meta-models* for building energy analysis.

The difference between conventional load/energy prediction and *meta-models* can be summarized from four aspects: input determination, data availability, data quality, and initial data. There are significant differences in how to specify input variables



(c) GBM-DL1 model for all the data



(d) GBM-DL1 model for 1/5 of the data set

Fig. 16 (continued)

between conventional load/energy prediction and *meta*-models. The conventional data-driven models can be regarded as ex-ante forecasts [54]. The predictors should be estimated in the ex-ante forecast in building energy prediction. As a result, the predictors that are hard to predict should be excluded. For instance, occupancy behaviors have important influences on building loads [55] but are difficult to accurately forecast due to the inherent high uncertainty. In contrast, the *meta*-models of building energy analysis can be considered as the ex-post forecast in which the predictors are known [54]. Hence, the variables influencing building energy (including occupant behavior) can be added to the *meta*-models of building energy. The second aspect is data availability between conventional load/energy prediction and *meta*-models. For conventional load/energy prediction, the data are hard to be changed for a specific building. In contrast, the extra new data are easily created using the physics-based building energy models for the *meta*-model construction. The effects of data availability would be discussed in the following paragraph. The third aspect is that the data quality from the conventional load or energy prediction of buildings should be carefully checked [56], whereas there are no data quality issues in the *meta*-models of building energy. The fourth aspect is about the initial data at the current time. For the conventional load/energy prediction, the initial data is the energy use measured at the current time. The 1-hour or 24-hour ahead prediction can be conducted based on this initial data. For the *meta*-models, there are no initial data since there are no measurement data. If energy use is regarded as an input variable, there are different ways to create this initial data. The initial

energy use can be regarded as zero if the HVAC system is turned off at a specific time. If the initial energy use is not zero, then the initial energy data can be estimated by running the first-day simulation several times to check the stability of energy results.

One of the differences between load/energy prediction and *meta*-model construction of buildings is the data availability as discussed in the previous paragraph. It is much easier to create data from engineering-based building energy models. Hence, the construction of time series *meta*-models should take advantage of this feature to implement the multi-step procedure. For instance, the model performance from ten (maybe more) machine learning models can be firstly assessed from a small number of data using cross-validation. This step is necessary because the construction of machine learning models is very time-consuming for deep learning models and some ensemble learning models. Moreover, the prediction time is also long for some autoregressive *meta*-models since it is almost impossible to use parallel computation because the energy used in the latter hour depends on the previous hours. The several learning models with better performance can be chosen from the preselection procedure to be further assessed using more large data. The method proposed here has a good sequential feature to be more suitable for the construction of reliable time series *meta*-models in buildings.

The choice of machine learning models for creating *meta*-models in buildings should be balanced between prediction capability and computational cost. In terms of prediction capability, the deep learning models and the ensemble models have good performance. The deep learning models are usually more time-

consuming and require more effort to tune parameters. The ensemble models are more flexible to create fast and reliable energy models. For stacking and bagging ensemble models, parallel computation can be easily implemented to expedite the process of creating building energy models. For the boosting models, it is hard to use parallel computing techniques due to their sequential nature. However, there are several boosting models, which can run fast, such as Cubist models. Overall, the boosting and stacking learning models are recommended to construct the time series building energy models.

The hourly *meta*-models developed from this research can be used for various purposes by replacing computationally expensive engineering-based models of buildings. These applications include the optimization of heating systems, demand response, model calibration, model predictive control, and sensitivity analysis. The models of building components and subsystems, especially HVAC systems, are often required to be calibrated based on hourly or sub-hourly data [9]. The hourly *meta*-models obtained in this paper are more easily combined with the ABC method [26] for the model calibration of building energy models, compared with the individual models at the different hour scales. Besides, the building hourly *meta*-model can also reduce the energy consumption of building operations while ensuring the thermal comfort of indoor occupants by optimizing multiple target parameters, such as HVAC set-point temperature, lamp, and equipment utilization rate [57,58]. The application of *meta*-models for building performance optimization has the advantage of significant time savings over physical simulation models.

4. Conclusion

This paper explores the construction of fast and accurate time-series *meta*-models to replace the engineering-based building energy model. The following conclusions can be obtained from this research.

(1) A novel method has been proposed to create machine learning-based time series *meta*-models for assessing building energy performance. The main steps include data preparation, model preselection, and data folding. The data preparation is to prepare the data suitable for time series analysis. The model selection can choose the learning models with good predictive performance and moderate computational cost from a number of machine learning models using a small amount of data. The data folding can assess the predictive performance of learning models chosen from the previous step to provide a robust evaluation of model performance.

(2) The distributed lag models perform well in creating time series *meta*-models of building energy. The predictive performance of autoregressive distributed lag models should be carefully assessed due to the accumulative errors from the estimation of energy use in previous hours.

(3) Computational time for the building energy *meta*-models should be carefully checked, especially for autoregressive models, because the computational time is the key advantage for *meta*-models to replace the engineering-based building energy models. The prediction of energy use should be used as input predictors in predicting the building energy use in the next hour for the autoregressive time series models. Then it is hard to apply parallel computing to expedite the computation in this case.

(4) The accuracy of time-series *meta*-models in building energy analysis does not necessarily increase with an increase in data size. Hence, it is necessary to check the stability of model accuracy. The data folding technique combined with variable importance can be used to provide robust analysis to this end.

(5) The ensemble machine learning methods perform well, including Cubist, GBM, and stacking, in terms of predictive capability

and computational cost. These methods can be used for creating reliable time-series building energy *meta*-models. The deep learning methods have good predictive performance at the expense of high computational cost and high tuning parameter efforts. For the optimal models for both heating and cooling energy prediction from this research, the R^2 values are above 0.99 and the CV(RMSE) (coefficient variation of root mean square error) values are less than 0.06.

(6) The method proposed here can be easily combined with the sequential sampling methods to provide more accurate *meta*-models by incrementally adding more data sets. These sequential features can make full use of the characteristics of *meta*-models, which can more easily add training data sets. Hence, the method proposed here can achieve a good balance between model accuracy and computational cost.

This research is only applied to an office building in Tianjin, China. More building types in various locations should be explored to assess time series features in creating time series *meta*-models for building energy assessment. Note that the method proposed in this research can be easily extended for different time steps of building energy analysis, such as sub-hourly, daily, and weekly. Another interesting area is to explore prediction intervals for the machine learning-based time series models of building energy. Moreover, the models developed in this analysis should be applied in model calibration, sensitivity analysis, optimization, and model predictive control at short-time intervals to evaluate their suitability in different contexts of building energy assessment.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This research was supported by the National Natural Science Foundation of China (No. 51778416) and the Key Projects of Philosophy and Social Sciences Research, Ministry of Education (China) "Research on Green Design in Sustainable Development" (No. 16JZDH014).

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